

Basic Concepts of Gene Cloning

Further Readings:

“Genome II” by TA Brown

“Gene Cloning” by TA Brown

Elements of Genetic Engineering

- Identification of target genes
- Isolation of target genes
- Amplification of target genes
- Study of the expression regulation of the target genes
- Functional analysis of target genes
- Modification of target genes
- Transformation of target genes

Basic Steps in Gene Cloning

- Cut out the target gene
- Ligate the target gene into a vector
- Transform the construct into a host
- Select the right clones in a host cell
- Propagate

How to Cut DNA at a Specific Site?

Types of Restriction Enzymes

Characteristics	Type I	Type II	Type III
Restriction and modification (methylation) activities	Single enzyme	Separate	Separate with a subunit in common
Protein structure	3 subunits	Simple	2 subunits
Requirements	ATP, Mg^{2+}	Mg^{2+}	ATP, Mg^{2+}
Recognition sites	No rotational symmetry	Rotational symmetry except type IIS	No rotational symmetry
Cutting sites	Possibly random, at least 1000 bp from the recognition sites	At or near (IIS) recognition sites	24-26 bp to 3' of recognition sites
Methylation sites	Recognition sites	Recognition sites	Recognition sites

Nomenclature of Restriction Enzymes

- The 1st letter (in capital and italics) = first initial of Genus name (from which the enzyme was isolated)
- The 2nd and 3rd (in italics) = the first 2 letters of the species name

e.g. *Hin* = *Haemophilus influenzae*

- The 4th letter (sometimes in italics) = strain or type
e.g. *Hind* = *Haemophilus influenzae* Rd

- The roman number followed is given to distinguish different restriction and modification system in the same strain

e.g. *HindIII*

Rare Cutters

- There are only a few enzymes that have recognition sites of 7 to 8 bp
- However, some genomic DNA molecules are deficient in certain motifs
- E.g. 5'-CG-3' is rare in human
 - *SmaI* (5'CC**CG**GG3') cuts every 78 kb
 - *BssHII* (5'G**CGCG**C3') every 390 kb
 - *NotI* (5'G**CGG**CC**GC**3') every 10Mb

Termini Generated by Restriction Endonucleases

- Cohesive ends

— 5'-overhang, e.g. *EcoRI*

5'..NNGAATTCNN..3'
3'..NNCTTAAGNN..5'

5'..NNG pAATTCNN..3'
3'..NNCTTAAp GNN..5'

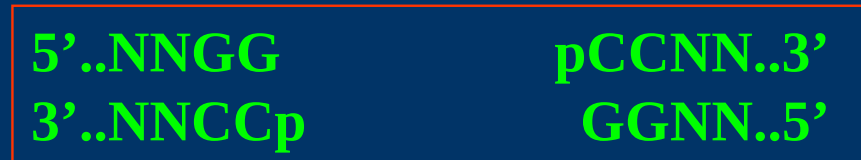
— 3'-overhang, e.g. *PstI*

5'..NNCTGCAGNN..3'
3'..NNGACGTCNN..5'

5'..NNCTGCA pGNN..3'
3'..NNGp ACGTCNN..5'

Termini Generated by Restriction Endonucleases

- Blunt ends; e.g. *HaeIII*



Isoschizomers

- Restriction enzymes that cleave within the same target sequences
 - e.g. *MboI* vs *Sau3AI*

↓
..NNGATCNN..
..NNCTAGNN..
↑

- e.g. *SmaI* vs *XmaI*

↓
..NNCCCGGGNN..
..NNGGGCCCNN..
↑

↓
..NNCCCGGGNN..
..NNGGGCCCNN..
↑

Isocaudamers

- Restriction enzymes that generate compatible ends

— e.g. *Bam*HI vs *Sau*3AI

↓
..NGGATCCN..
..NCCTAGGN..
↑

↓
..NNGATC NN..
..NNCTAG NN..
↑

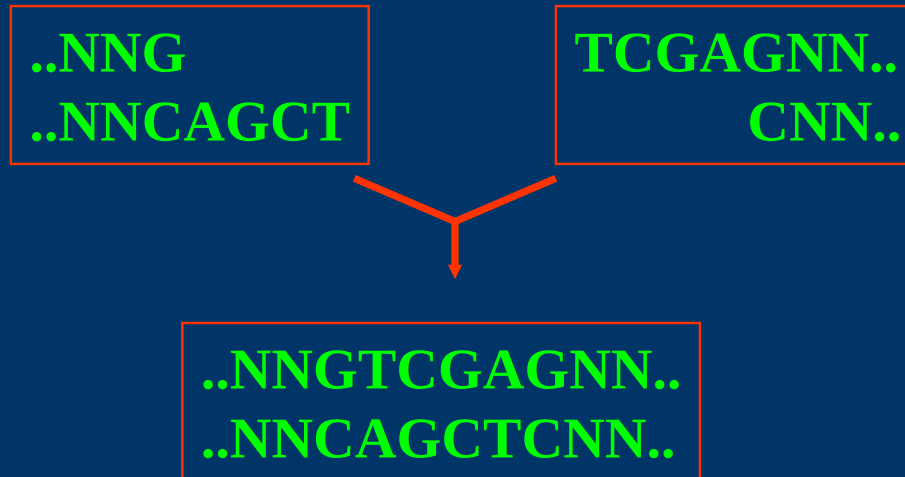
— e.g. *Sal*I vs *Xho*I

↓
..NNGTCGAC NN..
..NNCAGCTG NN..
↑

↓
..NNCTCGAG NN..
..NNGAGCTC NN..
↑

Compatible Ends

- Two restriction enzymes that generate the same sticky ends
 - e.g. *Sal*I vs *Xho*I



Additional Activities of Restriction Endonucleases

Star Activities

- Increasing glycerol concentration, changing pH, replacing Mg with Mn and reducing NaCl concentration may reduce the specificity of enzyme recognition.
- For examples, *EcoRI* cleaves GAATTC at pH 7.3 and 100 mM NaCl in the presence of 5 mM Mg, but raising the pH, lowering the NaCl concentration, substituting Mn for Mg or adding organic solvents all tend to reduce the specificity of cleavage to AATT

Nicking Activities

- Incubation at low temperature or in the presence of ethidium bromide may result in only single-strand cleavage

Cleavage of DNA/RNA Hybrids

- Several enzymes can cut DNA/RNA hybrids, e.g. *EcoRI*, *HindIII*, *Sall*, *MspI*, *HhaI*, *AluI*, *TaqI*, and *HaeIII*

Cleavage of Single-Stranded DNA

- Several enzymes can cut single-stranded DNA with some degree of efficiency and specificity, e.g. *HaeIII*, *HhaI*, *SfaI*, *MboI*, *HinfI*, *HpaI*, *PstI*, *BluI*, and *AvaI*

Methylation in Commonly Used *E. coli* Host Strains that Will Affect Restriction Digestion

dam methylase

- Introduces methyl groups at the N⁶ position of A in the sequence 5'GATC 3'
- To check methylation, use the enzyme pair *Sau3AI* (cutting not affected) and *MboI* (cutting inhibited)
- To cleave prokaryotic DNA at every possible site with *ClaI*, *XbaI*, *TaqI*, *MboII*, or *HphI*, or to cleave it at all with *BclI*, DNA must be prepared from *dam⁻ E. coli*.

dcm methylase

- Introduces methyl groups at the C⁵ position of the internal C in the sequences 5'CCAGG3' or 5'CCTGG3'
- *EcoRII* is one commonly used enzyme affected by dcm methylation

Basic Steps in Gene Cloning

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How to Ligate DNA Fragment to a Vector?

DNA Ligases

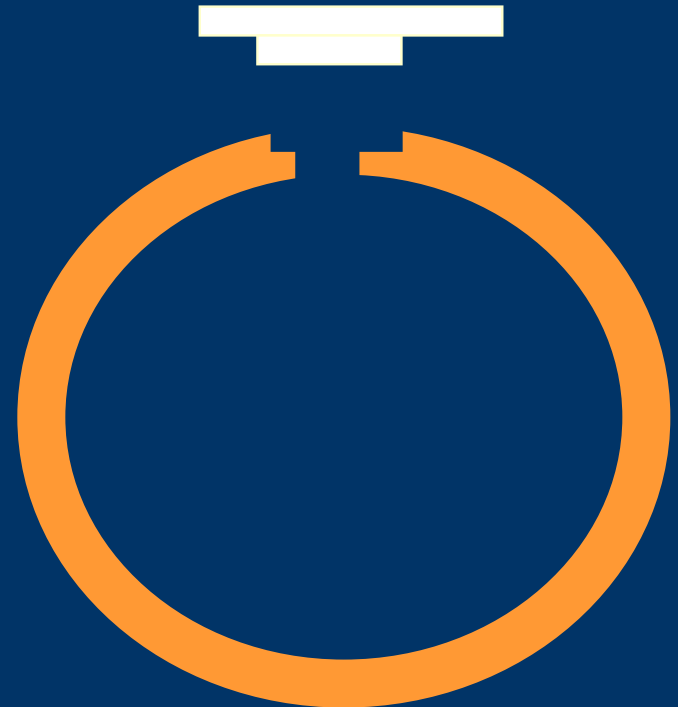
Type	Energy source	Used for:
<i>E. coli</i> ligase	NAD (turned into AMP and NMN)	Sticky ends
T7 ligase	ATP (turned into AMP and PPi)	Sticky ends
T4 ligase	ATP (turned into AMP and PPi)	Both sticky and blunt ends

Cloning with Compatible Ends

Non-directional

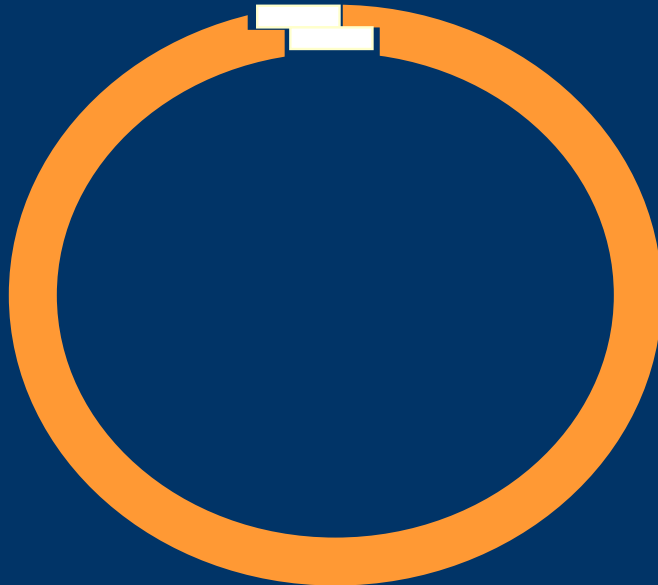


Directional

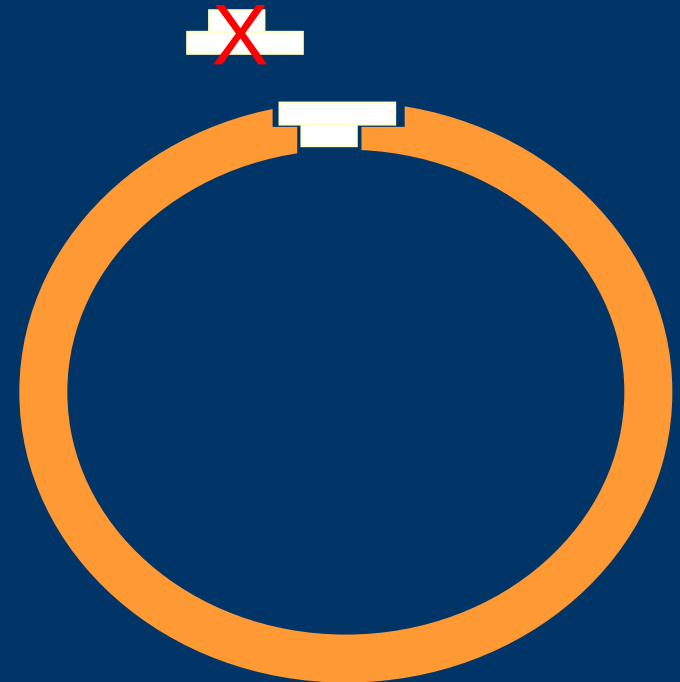


Cloning with Compatible Ends

Non-directional



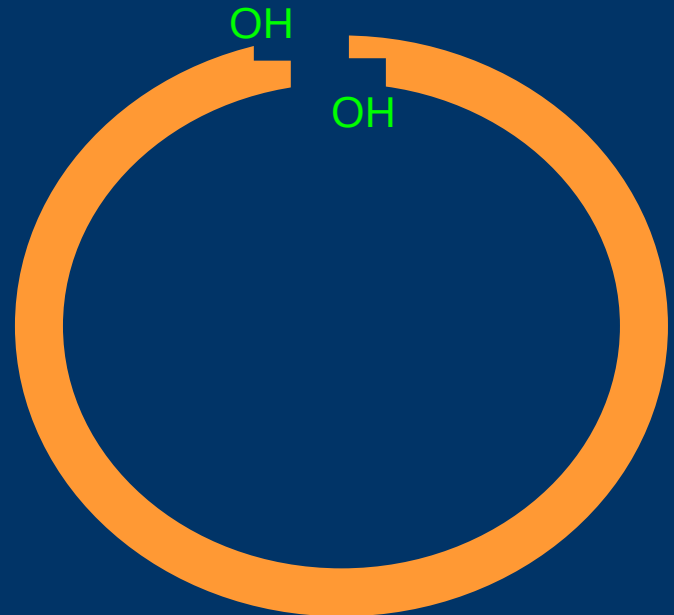
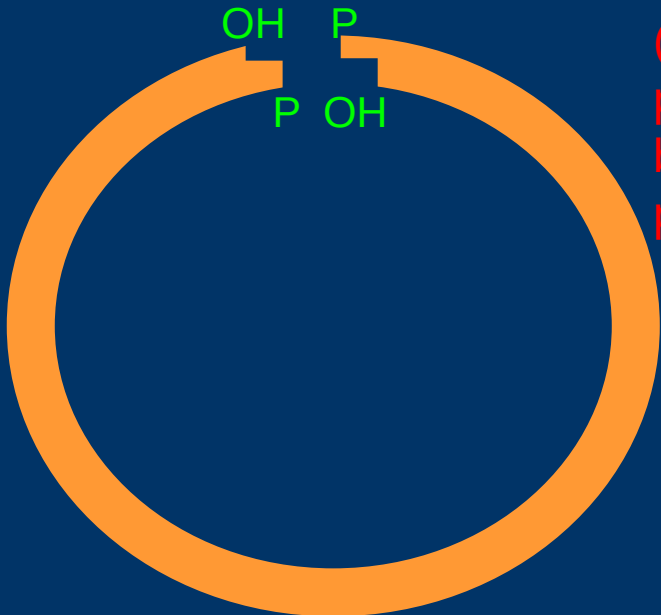
Directional



Preventing Self-Ligation of Vector



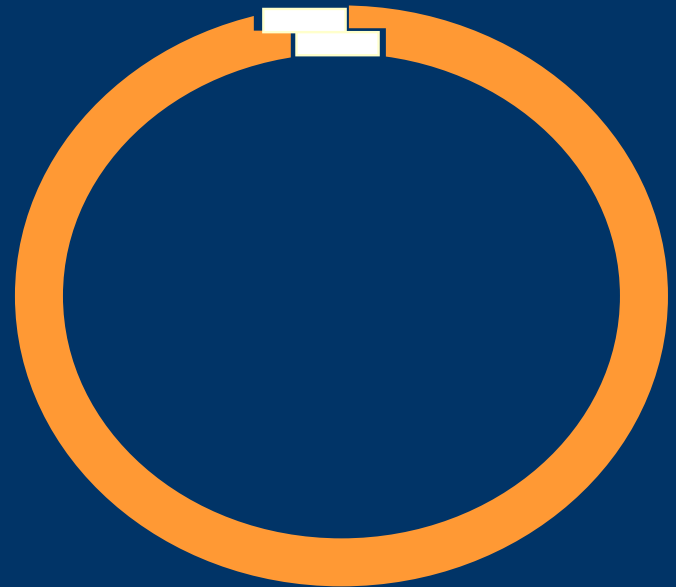
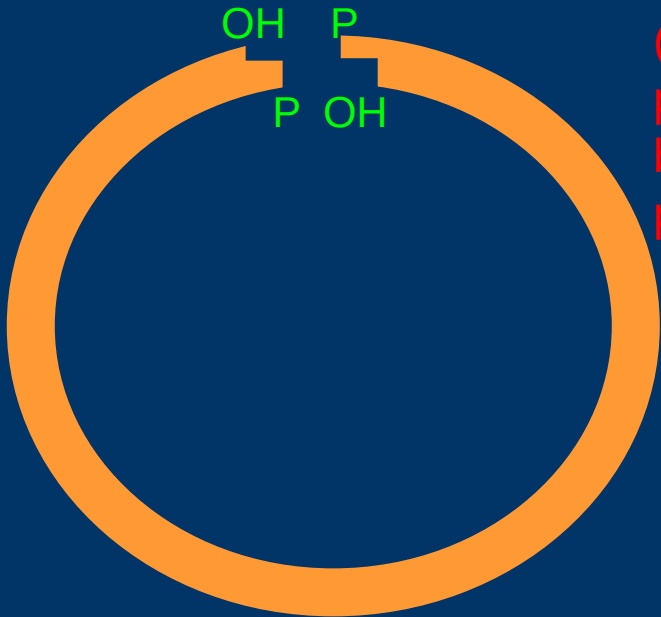
Dephosphorylation
by phosphatase
(e.g. calf intestine
phosphatase,
bacterial alkaline
phosphatase)



Preventing Self-Ligation of Vector



Dephosphorylation
by phosphatase
(e.g. calf intestine
phosphatase,
bacterial alkaline
phosphatase)



Cloning with Blunt Ends

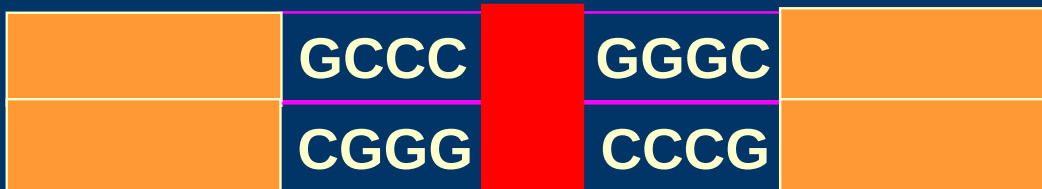
- Use T4 DNA ligase directly - low efficiency
- to increase efficiency: use small volume; increase insert to vector ratio; add hexamine cobalt chloride

Cloning with Blunt Ends

- PCR-Script (Stratagene; Add *SrfI* and Ligase at the same time)



**Self-Ligated Vector
is subject to *SrfI***



***SrfI* site lost after
DNA insertion**

Cloning with Blunt Ends

- Use terminal transferase

Insert



Partial digestion by 5' specific
exonuclease, e.g. λ exonuclease



+ terminal transferase and dATP



Vector



Partial digestion by 5' specific
exonuclease, e.g. λ exonuclease



+ terminal transferase and dTTP



Ligation

Cloning with Blunt Ends

- Use linkers

Blunt-ended insert



Linker (e.g. *Bam*HI)



Ligate to linker



Cut with *Bam*HI



Ligate to vector cut with *Bam*HI



Cloning with Incompatible Ends

- Use adapters

....NNG
....NNCAGCT

GATCCNN..NNG
GNN..NNCCTAG

TCGACNN....
GNN....

....NNGTCGAC...GGATCCNN..NNGGATCC...GTCGACNN....
....NNCAGCTG...CCTAGGNN..NNCCTAGG...CAGCTGNN....

Cloning with Incompatible Ends

- Converts ends into blunt ends

Cut with *Bam*HI (5' overhang)



+ Klenow; DNA polymerase I
or T4 DNA polymerase
+ dNTPs
(Fill-in)



Cut with *Pst*I (3' overhang)



+ T4 DNA polymerase
+ dNTPs
(Chew-back)



Blunt end ligation



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Cloning Vectors in *E. coli*

Plasmids

Phages

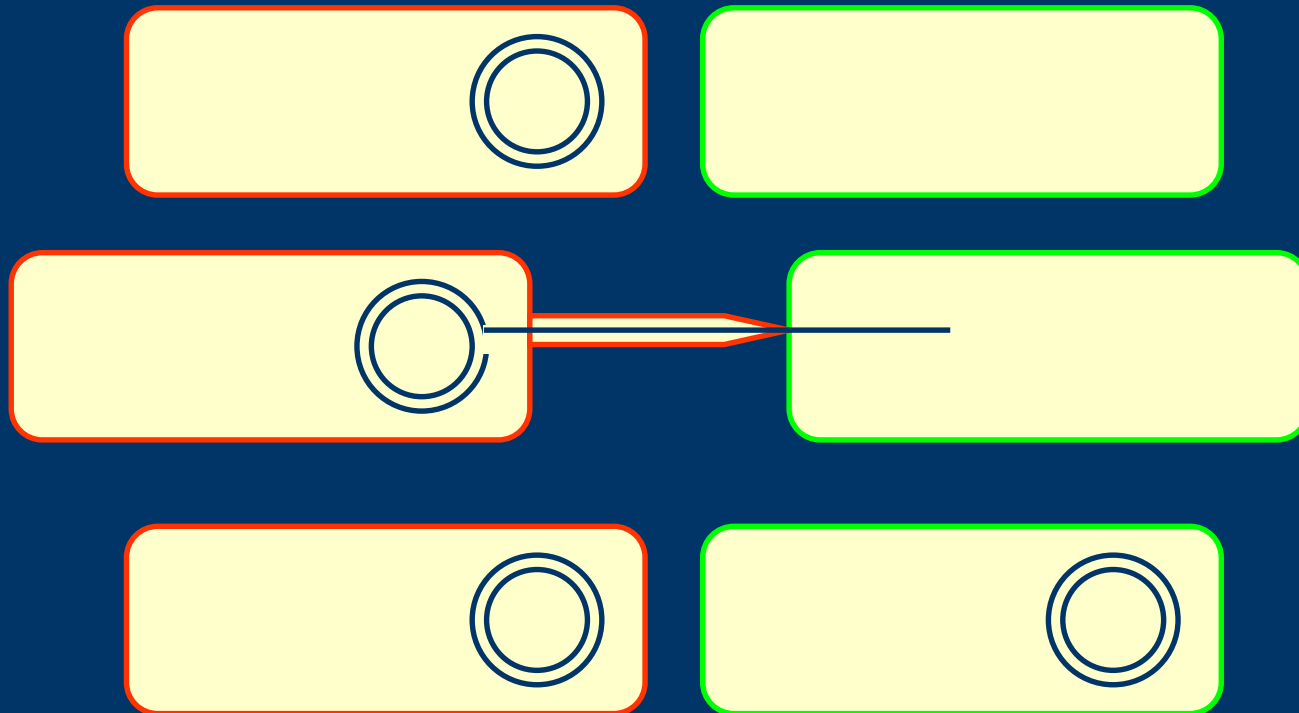
Plasmid Classification

- F; fertility, conjugal transfer of DNA
- R; resistance to antibacterial agents
- Col; produce colicins that kill other bacteria
- Degradative; metabolize unusual molecules
- Virulence; confer pathogenicity on the host bacterium

Plasmid Conjugation

Donor (F^+)

Recipient (F^-)



Compatibility of Plasmids

- The ability of two different plasmids to coexist in the same host in the absence of selection pressure
- Over 30 incompatibility groups found in *E. coli*, e.g. P, Q, W, etc.

Types of Plasmid Vector

- Basic cloning vectors
- Shuttle vectors
- Phagemids
- RNA expression vectors
- Protein expression vectors
- Promoter probe vectors
- Cosmids
- Cloning vectors for blunt-end ligation products
- Mutagenesis vectors

Basic Cloning Vectors:

Replication origin
Selectable markers

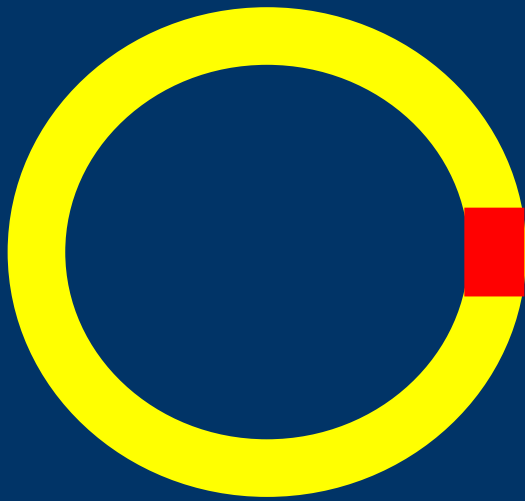
Selectable Markers for Plasmid Transformation in *E. coli*

Marker	Source	Gene product	Phenotype
amp	Tn3	β -Lactamase	Ampicillin resistance
cml or cat	Tn9	Acetyltransferase	Chloramphenicol resistance
kan	Tn5	Phosphotransferase	Kanamycin resistance
neo	Tn903	Phosphotransferase	Kanamycin resistance
tet	Tn10 / pSC101	Membrane protein	Tetracycline resistance

Selectable Markers for Plasmid Transformation in Yeast

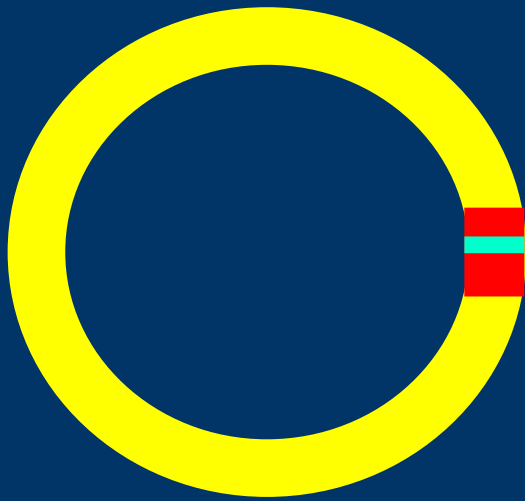
- Use metabolic markers mainly
- Transform into yeast strain defective in one or more of these metabolic markers
- e.g. ADE, HIS, LEU, MET, TRP, URA

Select for Recombinants (by Insertional Inactivation)



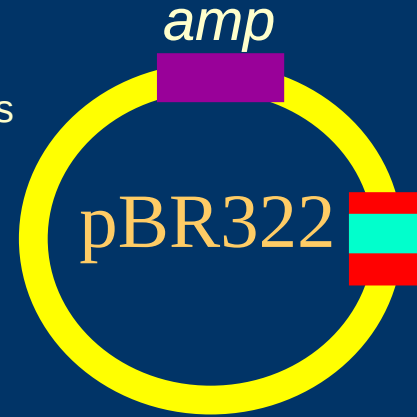
Gene Product
(e.g. Antibiotic
resistance, LacZ
function)

Select for Recombinants (by Insertional Inactivation)



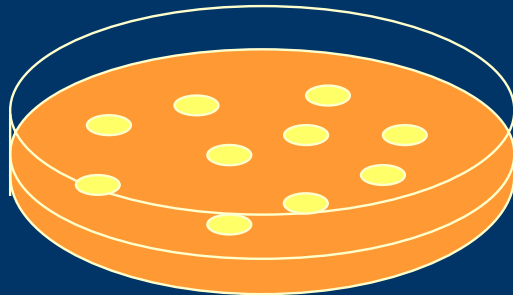
No Gene Product

Transform
into a Amp^s
bacteria



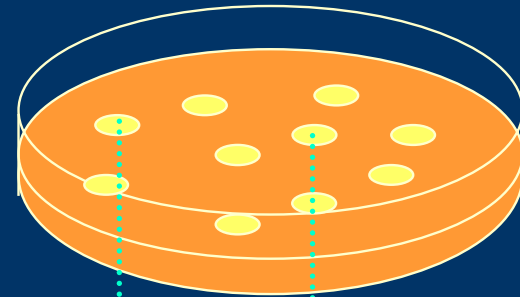
Inserted
with a DNA
fragment

Select on
ampicillin-
containing
media

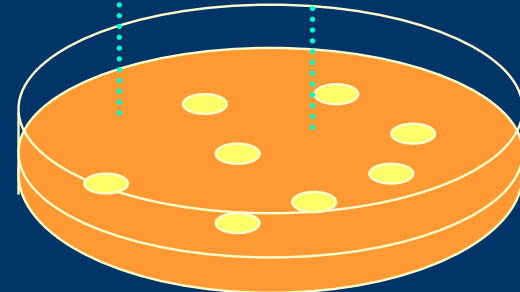


Master plate

Replica
plating

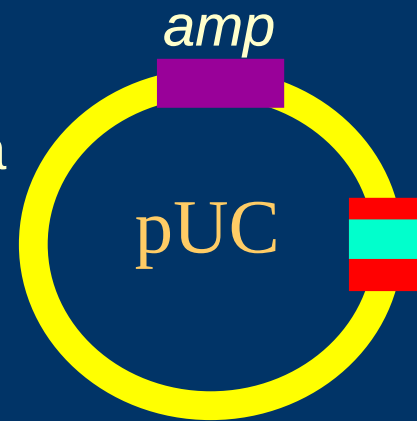


Amp
plate



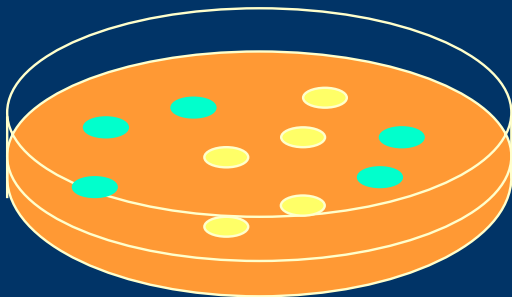
Tet
plate

Transform
into a Δ M15
lacZ bacteria



lacZ'
Inserted
with a DNA
fragment

Select on ampicillin-
containing media with
IPTG and X-Gal



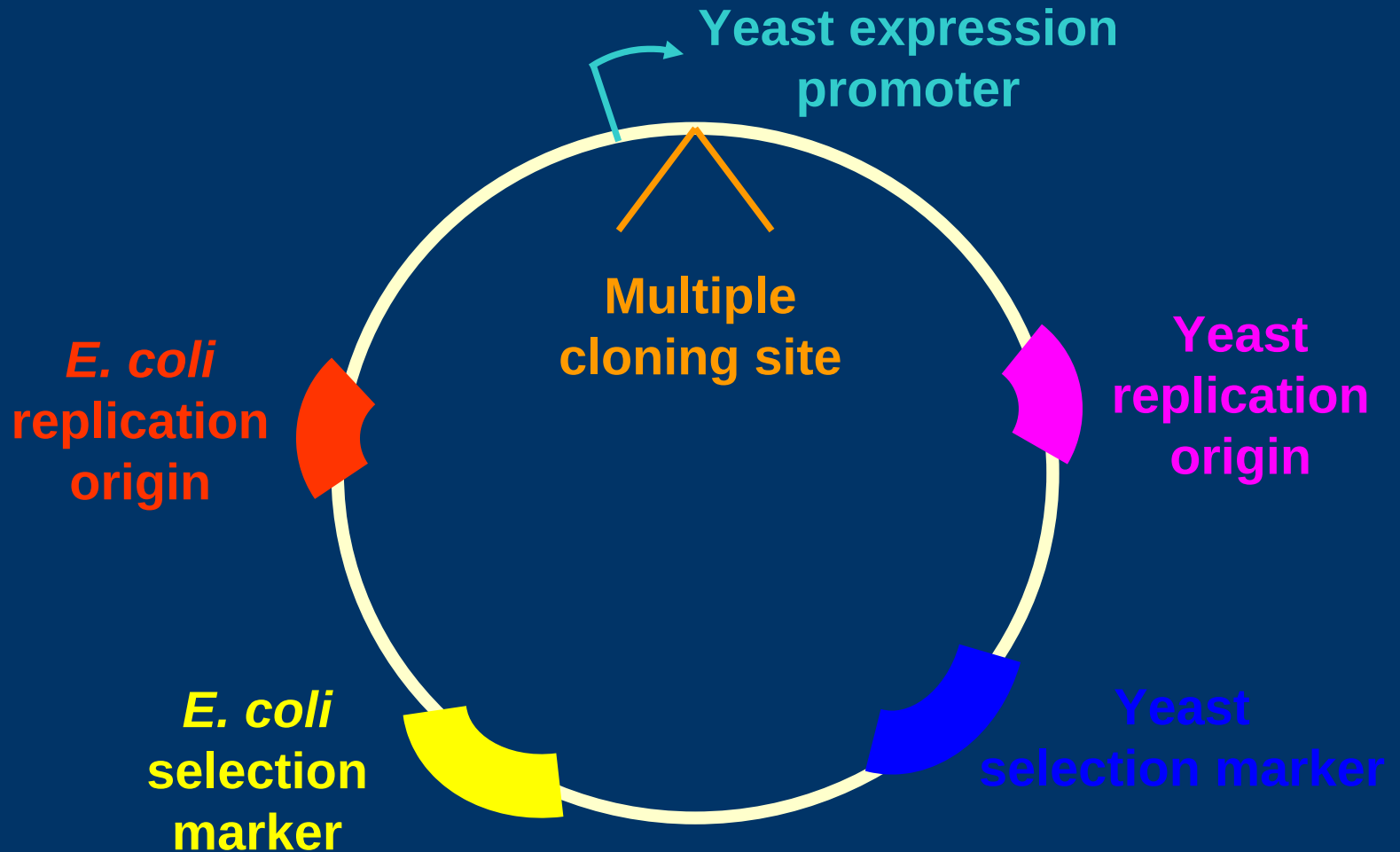
LacZ α Complementation

- *lacZ'* encodes α subunit of LacZ
- The Δ M15 *lacZ* encodes β subunit LacZ
- IPTG induces *lac* promoter to express the *lac* operon
- X-Gal turned blue by LacZ
- White colonies contain DNA inserts

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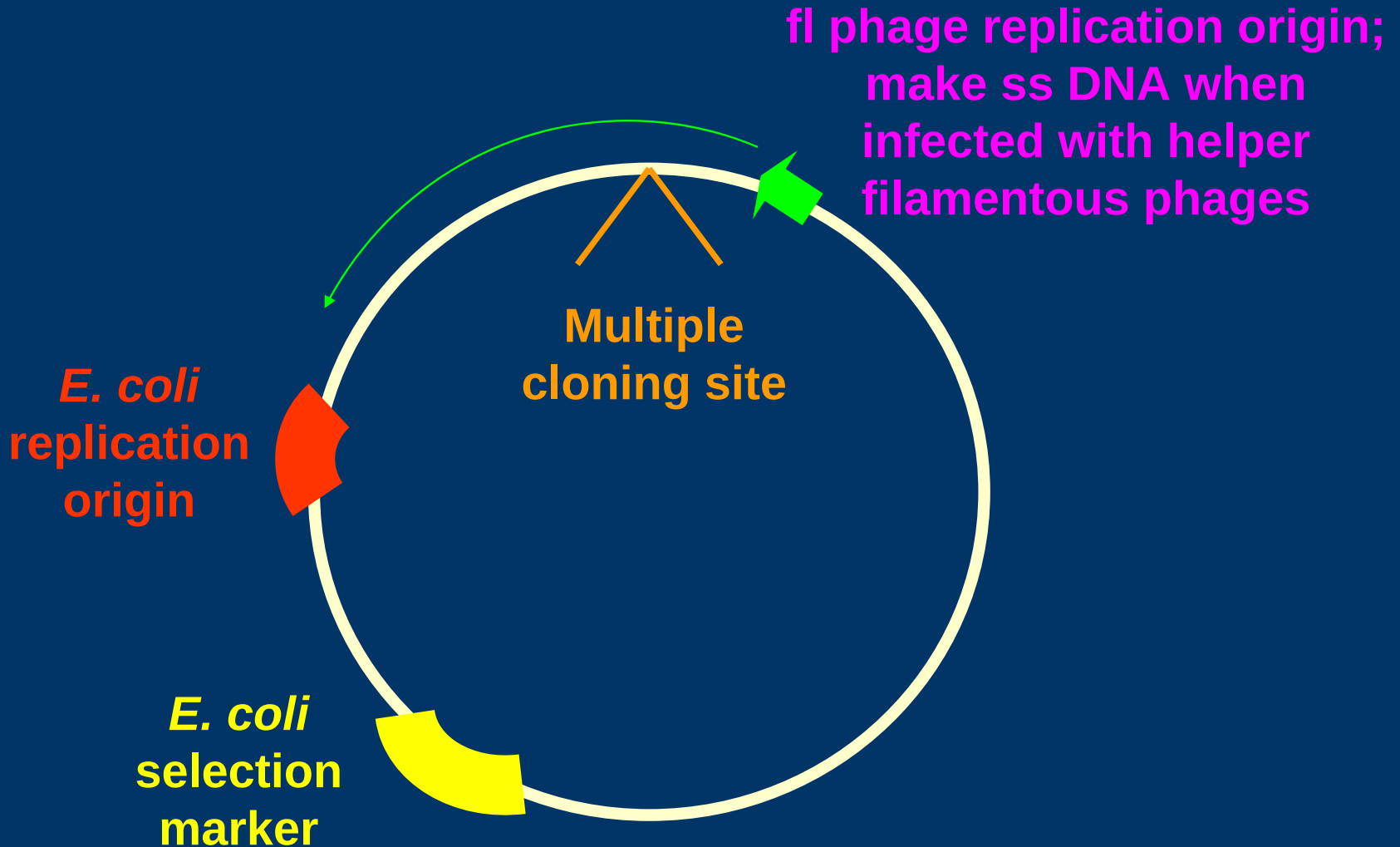
A Typical Yeast-*E. coli* Shuttle Vector (e.g. YES vector)



Types of Plasmid Vector

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Phagemid (e.g. pBluescript form Strategene)



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Promoters for RNA Synthesis in *E. coli* plasmid vector

- T3 promoter from T3 bacteriophage
- T7 promoter from T7 bacteriophage
- SP6 promoter from SP6 bacteriophage

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Promoters for Protein Expression in *E. coli* Cells

Promoter	Source	Induction
<i>lac</i>	<i>E. coli lac</i> operon	IPTG
<i>lac-tac</i>	Hybrid	IPTG
P_L	λ phage	clts857
P_R	λ phage	clts857
<i>tac</i>	<i>trp-lac</i> hybrid	IPTG
<i>trc</i>	<i>trp-lac</i> hybrid	IPTG
<i>trp</i>	<i>E. coli trp</i> operon	IAA

(Also need Shine-Dalgarno sequence AGGAGG for translation)

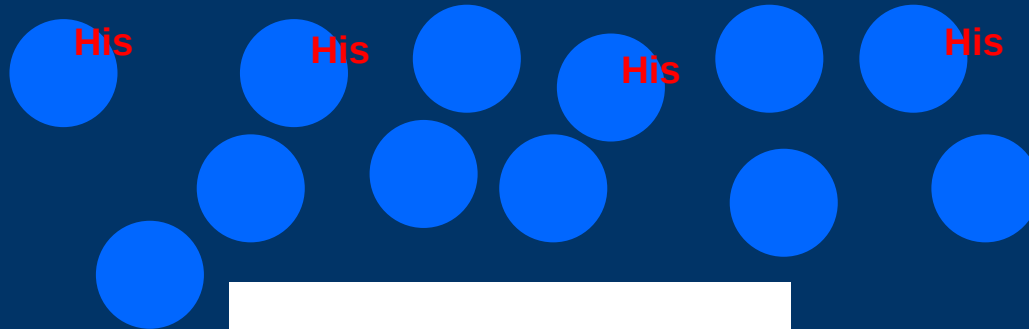
e.g. His-Tag pET System (Novagen)

Inducible Promoter

Target Protein

EK Cleavage Site

His Tag



**Ni Affinity
Column
(bind to a
chain of His
by chelation)**

Inducible Promoter

Target Protein

EK Cleavage Site

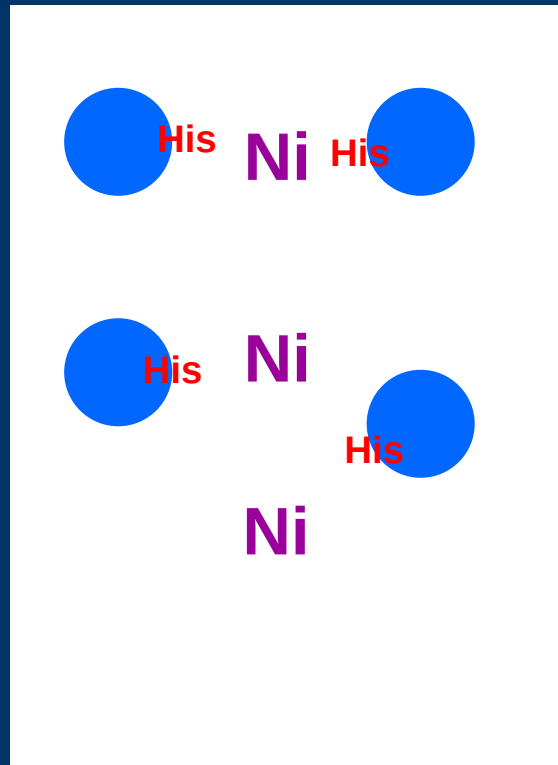
His Tag

Inducible Promoter

Target Protein

EK Cleavage Site

His Tag



**Ni Affinity
Column
(only retains
His-tagged
proteins)**

Inducible Promoter

Target Protein

EK Cleavage Site

His Tag



**Wash out
the His-Tagged
target proteins
by imidazole**

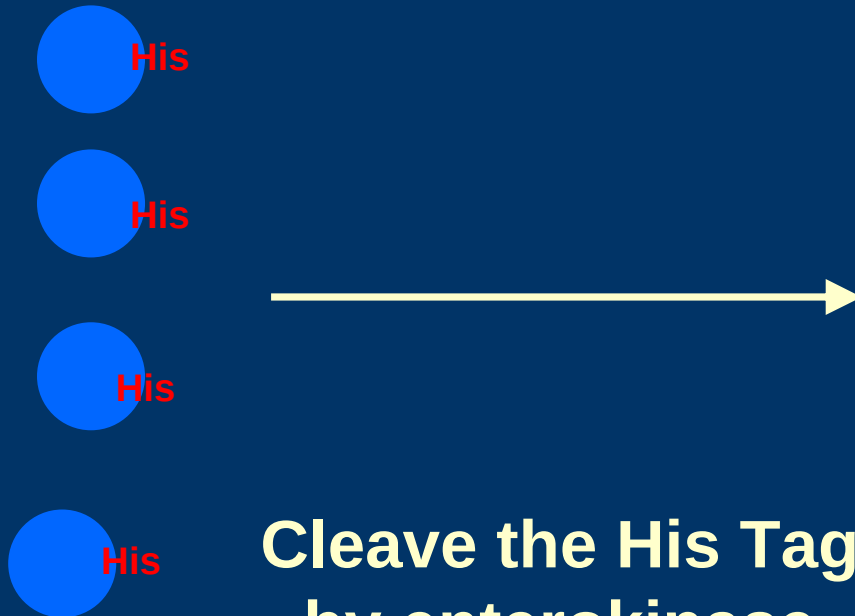


Inducible Promoter

Target Protein

EK Cleavage Site

His Tag



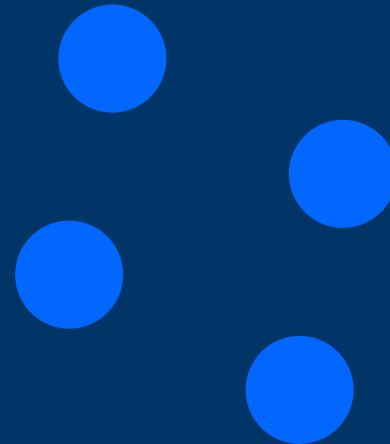
**Cleave the His Tag
by enterokinase**

Inducible Promoter

Target Protein

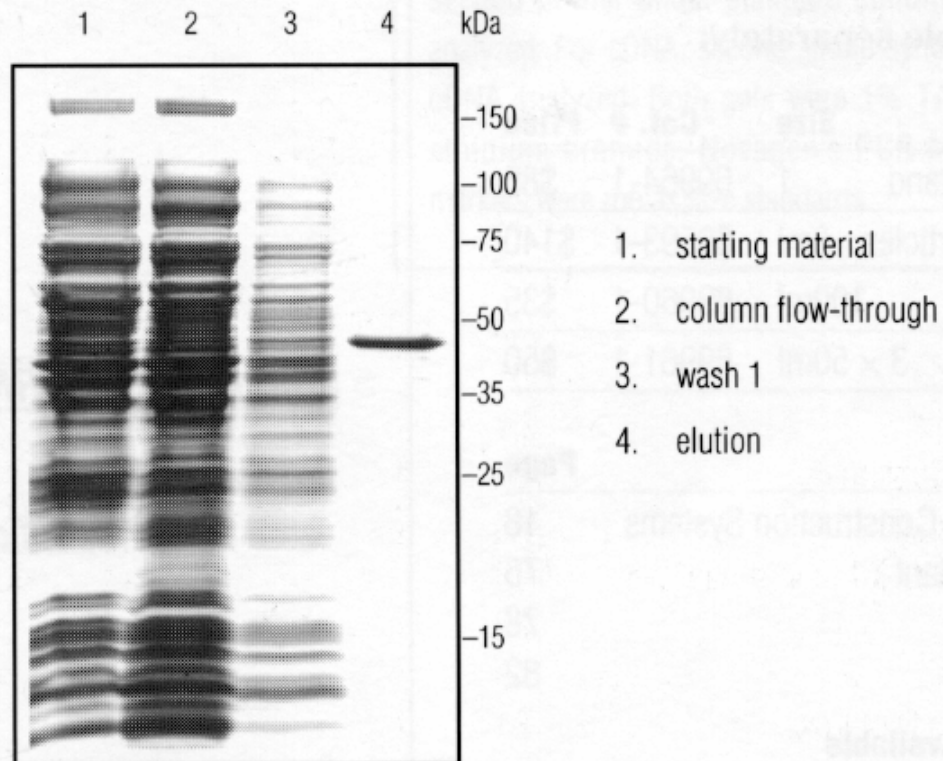
EK Cleavage Site

His Tag



**Purified
Products**

One-Step Affinity Purification of a Poorly Expressed Protein on His•Bind Resin

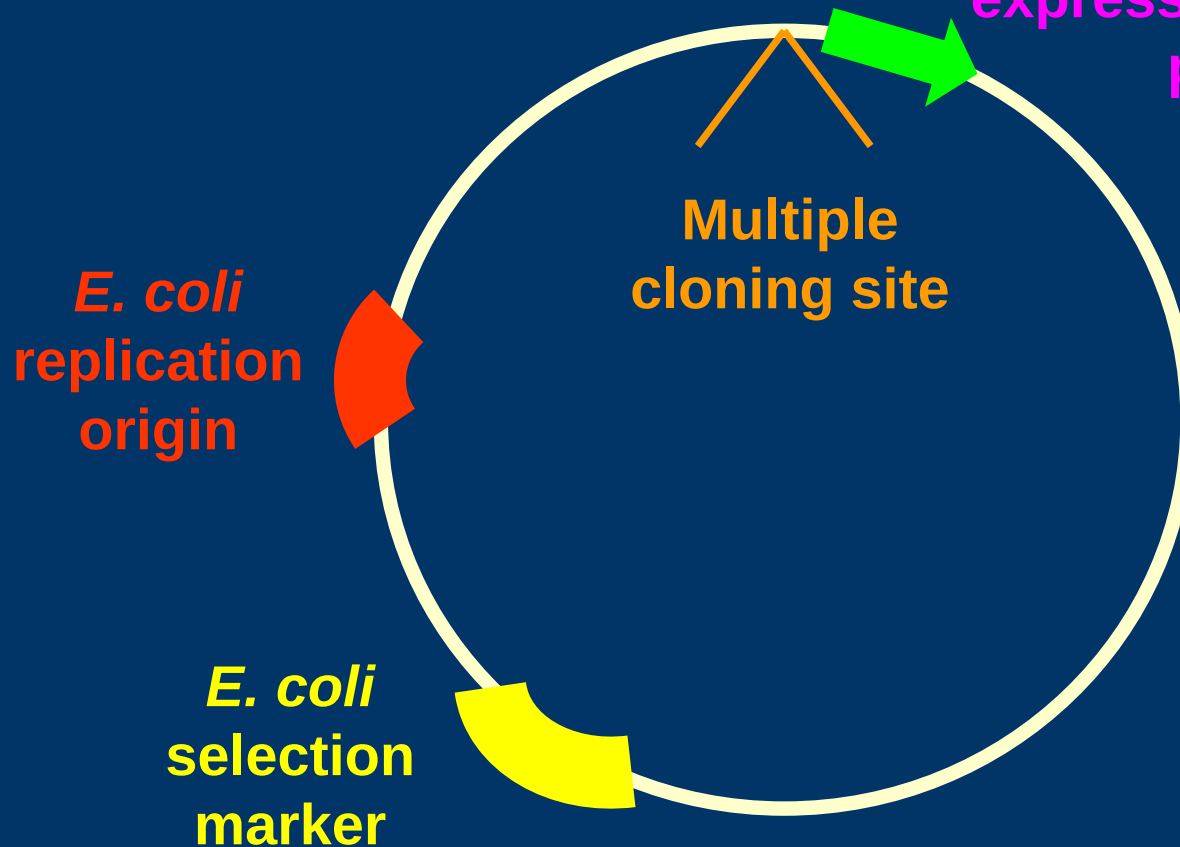


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e.g. pKK232-8

Contains a reporter gene
(e.g. CAT) to study the
expression of the cloned
promoters



In Vitro Reporter Assays

- Chloramphenicol acetyltransferase (CAT)
- Firefly luciferase
- Beta-galactosidase (LacZ)
- Secreted alkaline phosphatase (SEAP)
- Human growth hormone (hGH)
- Beta-glucuronidase (GUS)

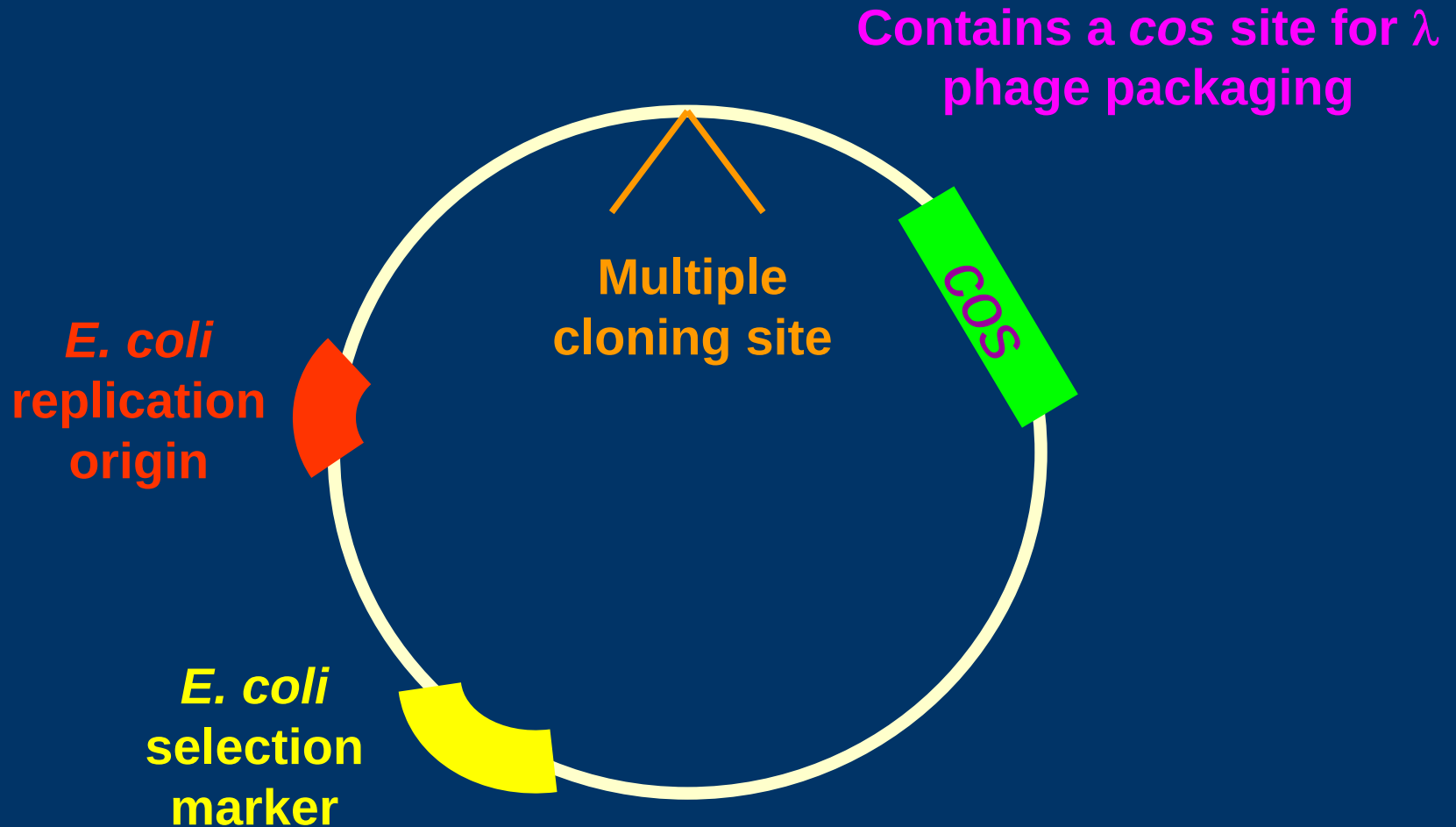
In Vivo Reporter Assays

- Green fluorescent protein (GFP)
- Firefly luciferase
- Beta-galactosidase (LacZ)
- Beta-glucuronidase (GUS)

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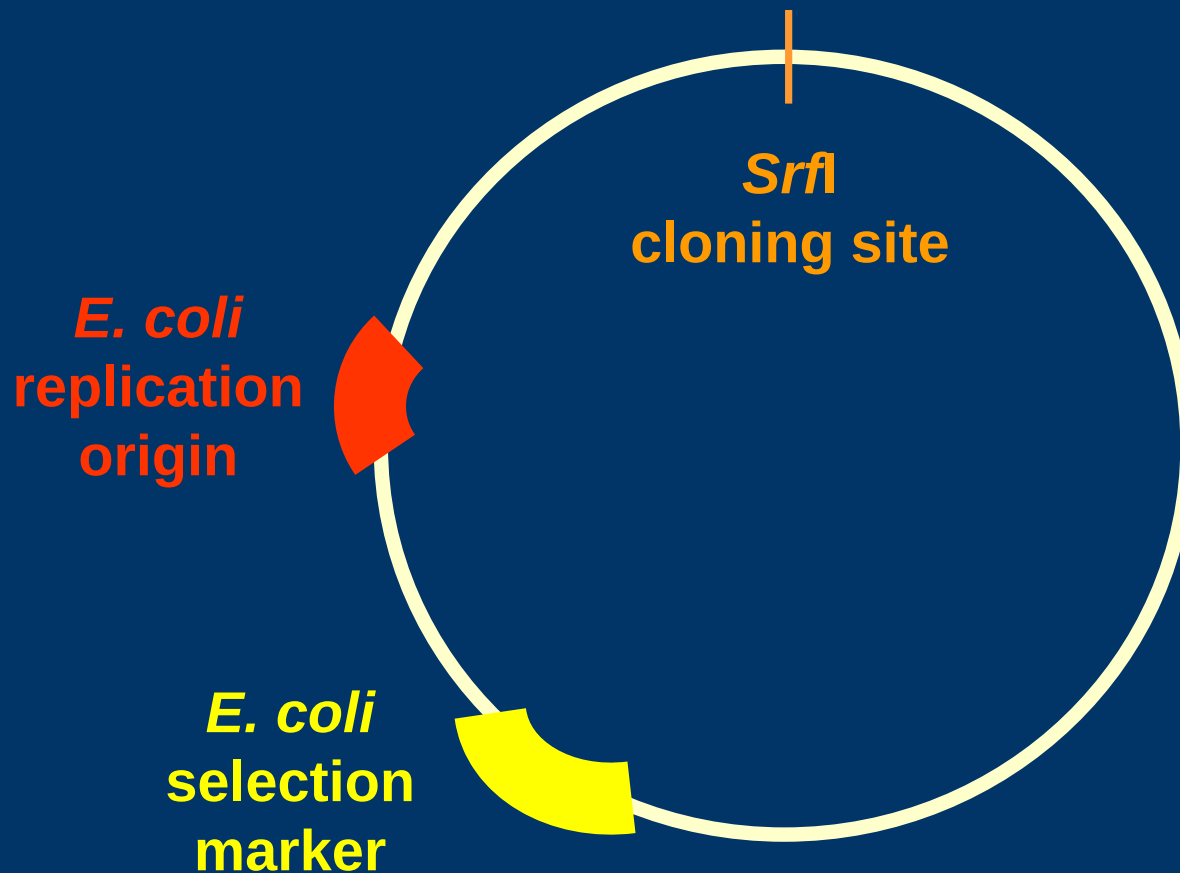
e.g. pJB8



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e.g. pCR Script Direct

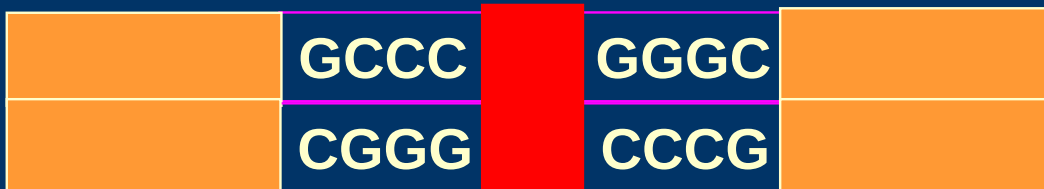


Cloning with Blunt Ends

- PCR Script (Add *SrfI* and Ligase at the same time)



**Self-Ligated Vector
is subject to *SrfI***

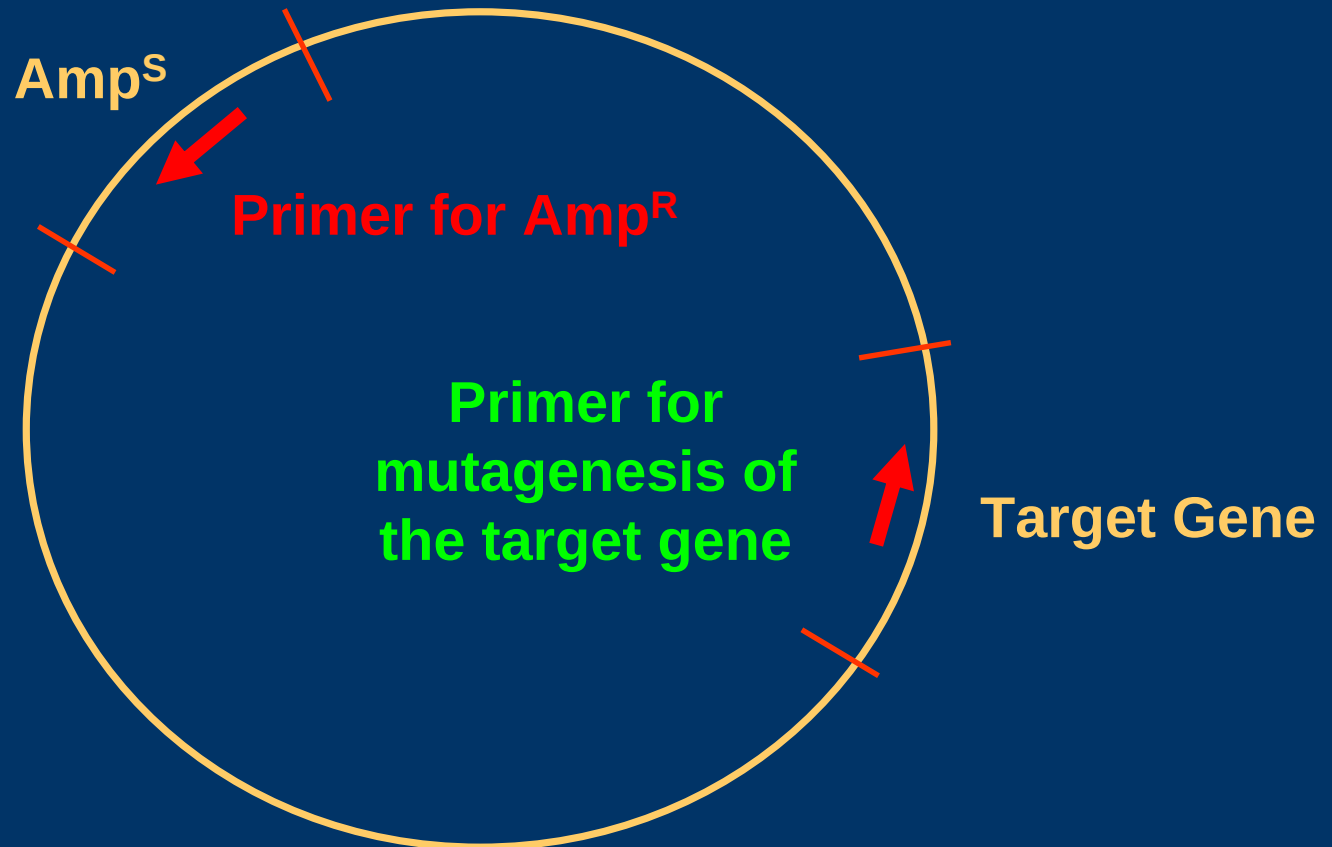


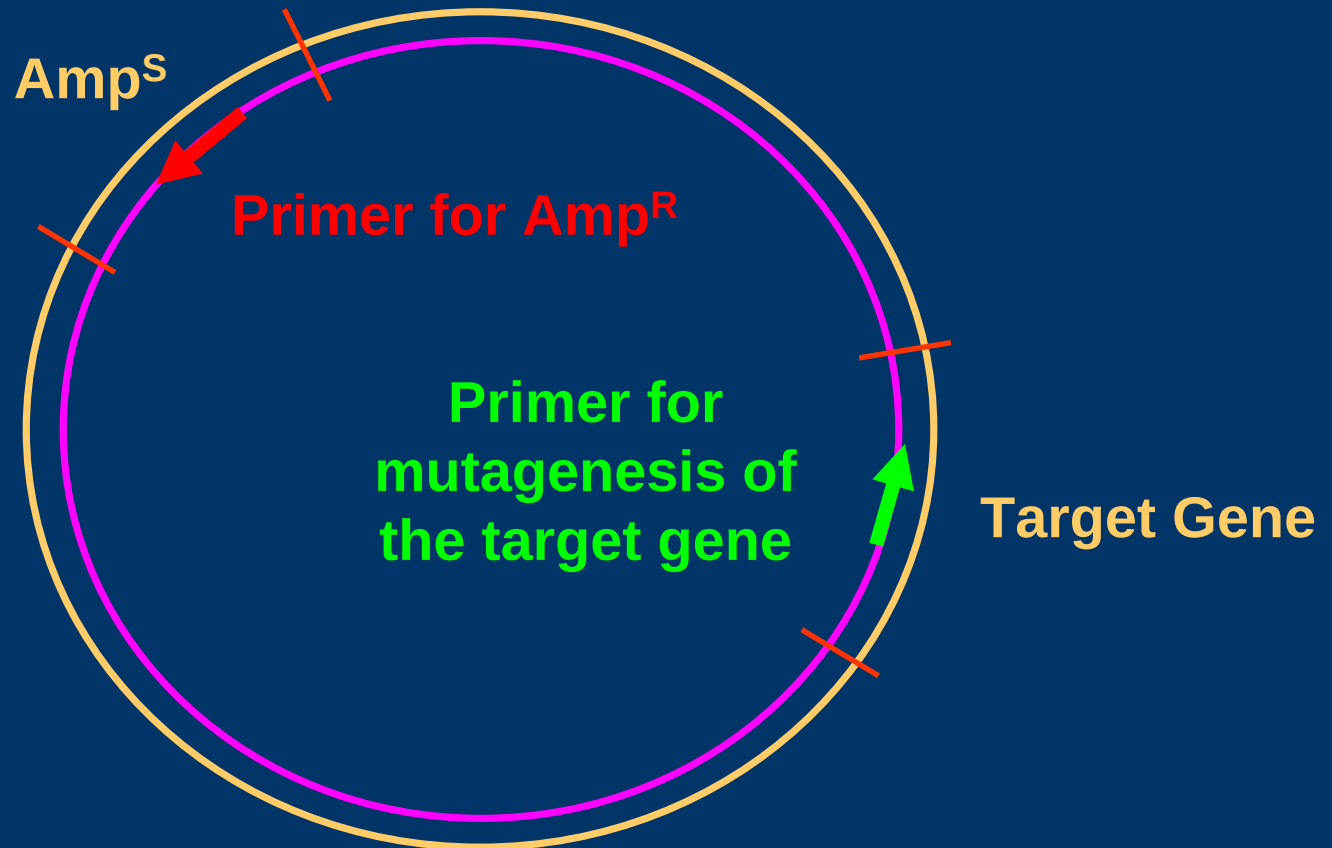
***SrfI* site lost after
DNA insertion**

Types of Plasmid Vector

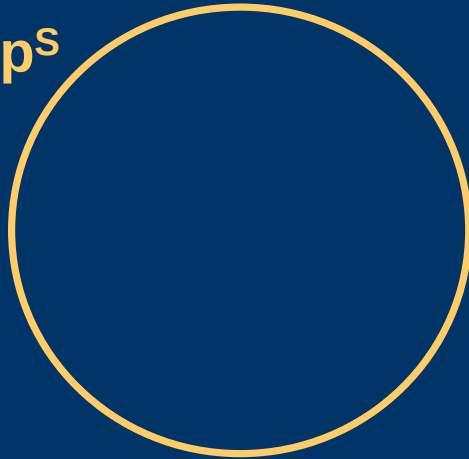
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e.g. pALTER-1



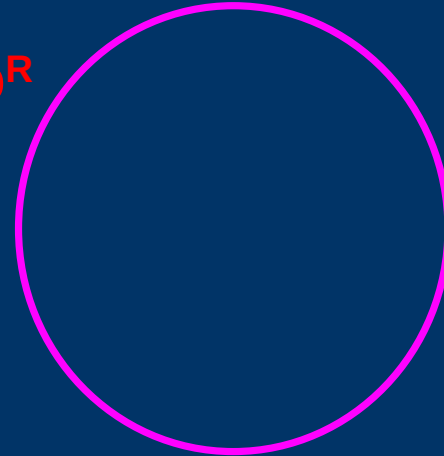


Amp^S



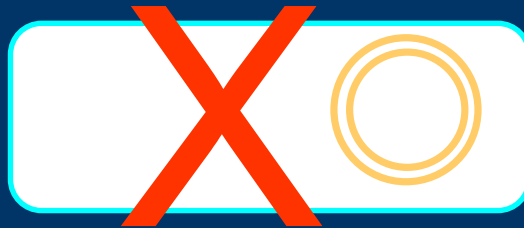
**Original
Gene**

Amp^R



**Mutated
Gene**

Transform *E. coli* and
select on ampicillin
containing media



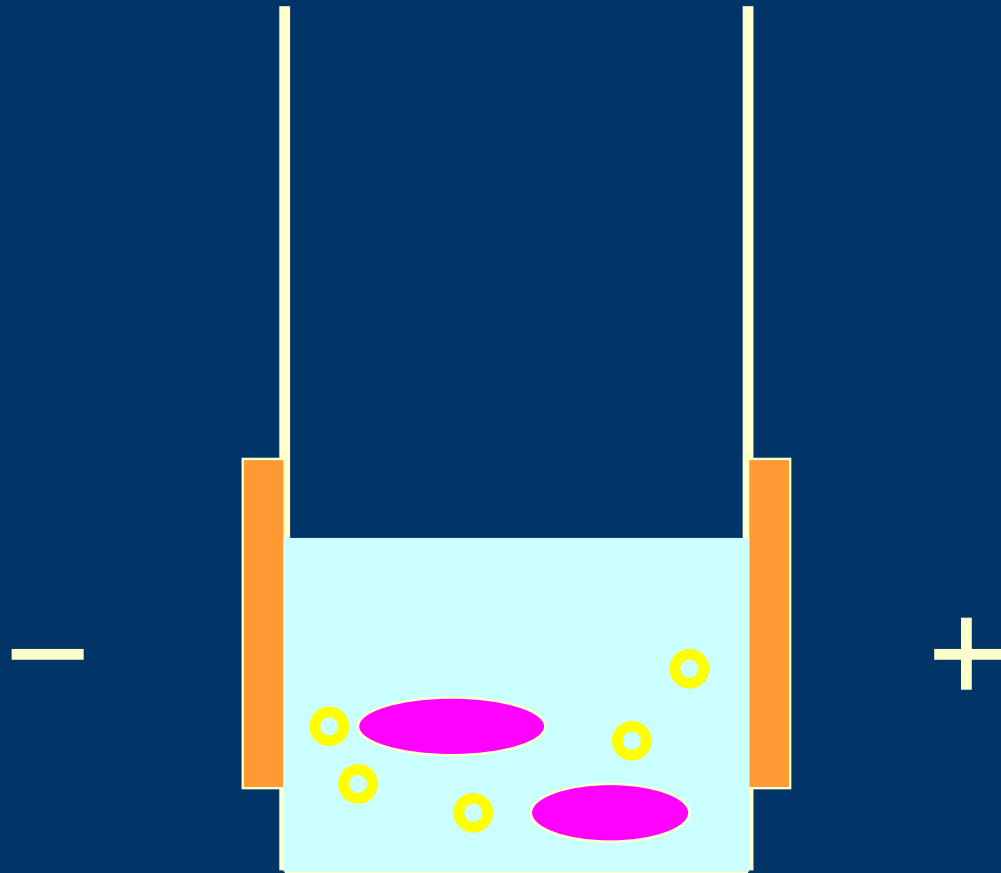
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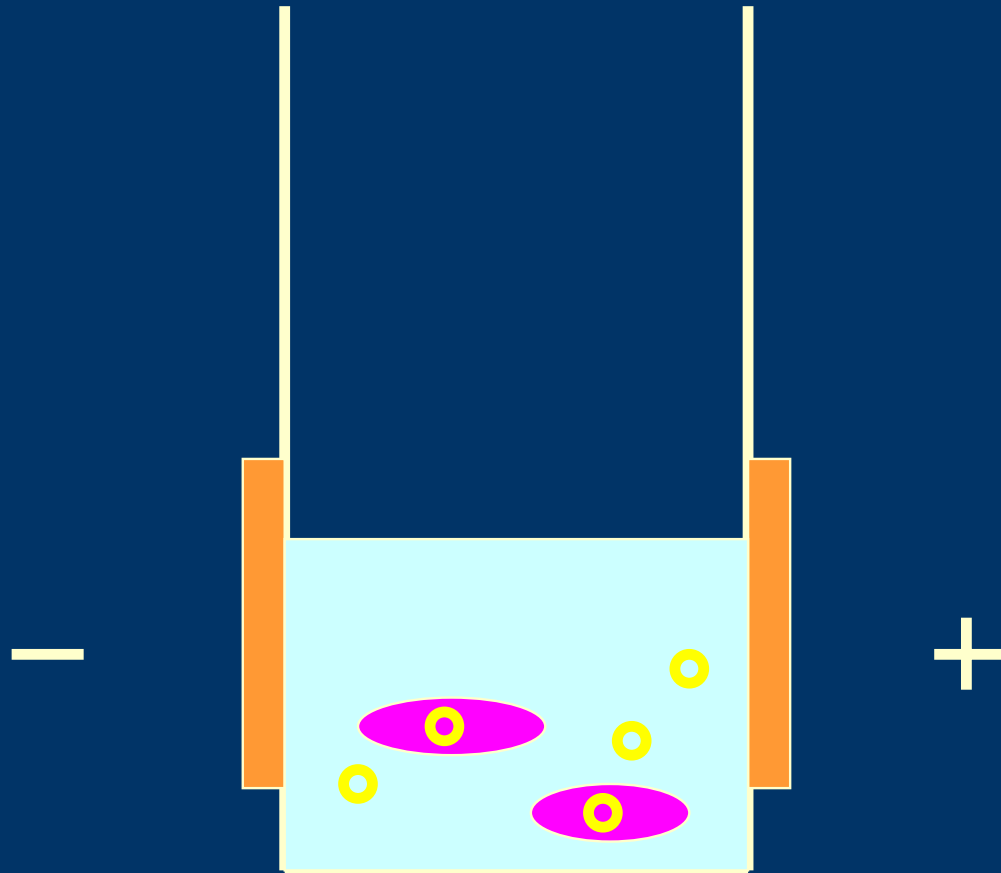
Plasmid Transformation into *E. coli*

- Calcium chloride method
 - treat cells with CaCl_2
 - heat pulse (42°C for 2'; 30°C for 10' in case of temperature sensitive bacteria)
- Electroporation
 - rinse cells thoroughly in deionized water
 - set the right conditions (e.g. 2.5 kV, 25 μF , 200 Ohm)

Transformation into Host by Electroporation



Transformation into Host by Electroporation



Consideration for the Host Strains

- Nonrestricting strains; *E. coli* K has at least 3 different methylation-dependent restriction systems; e.g. *mrr*⁻, *mcrA*⁻, *mcrB*⁻
- Minimize recombination; e.g. *recA*⁻
- Minimize protease activity; e.g. *lon*⁻

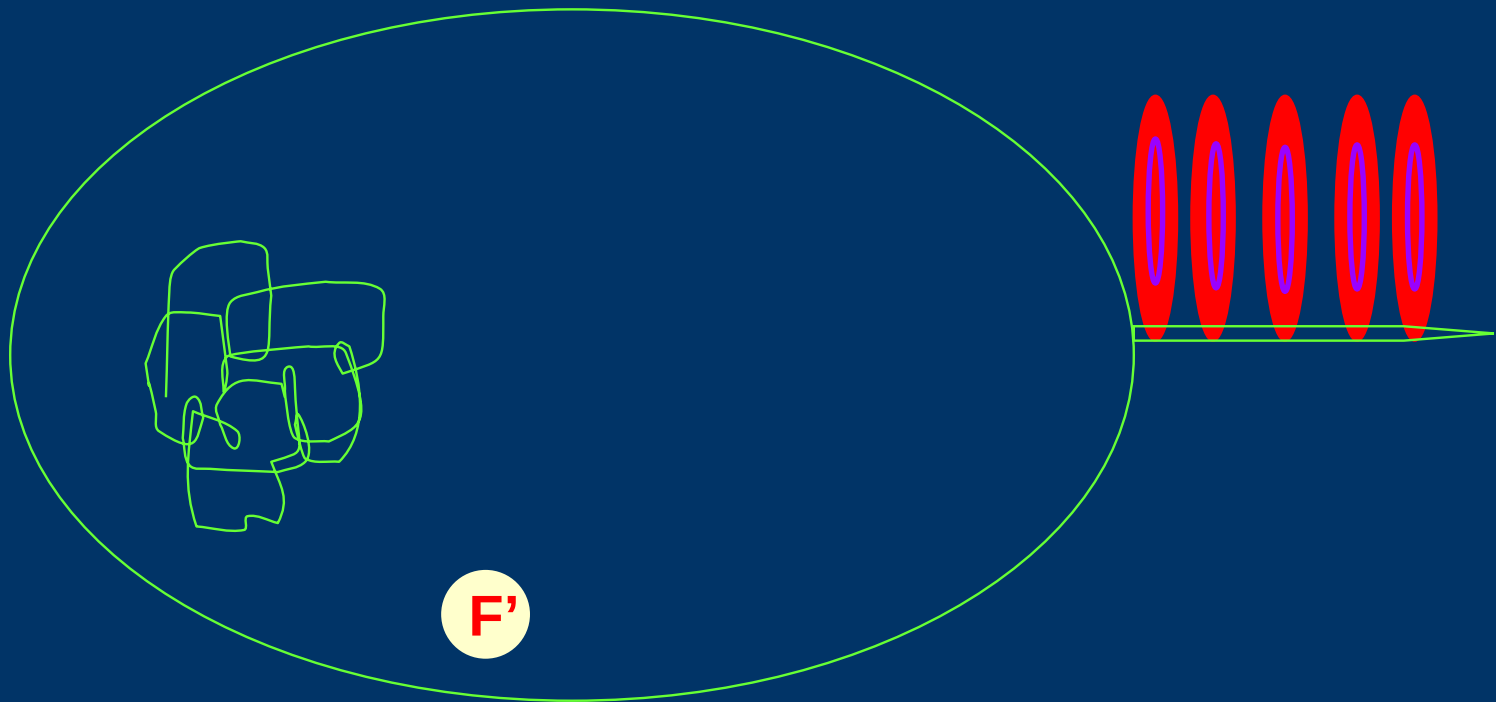
Phage Vectors Commonly Used in *E. coli*

λ phage

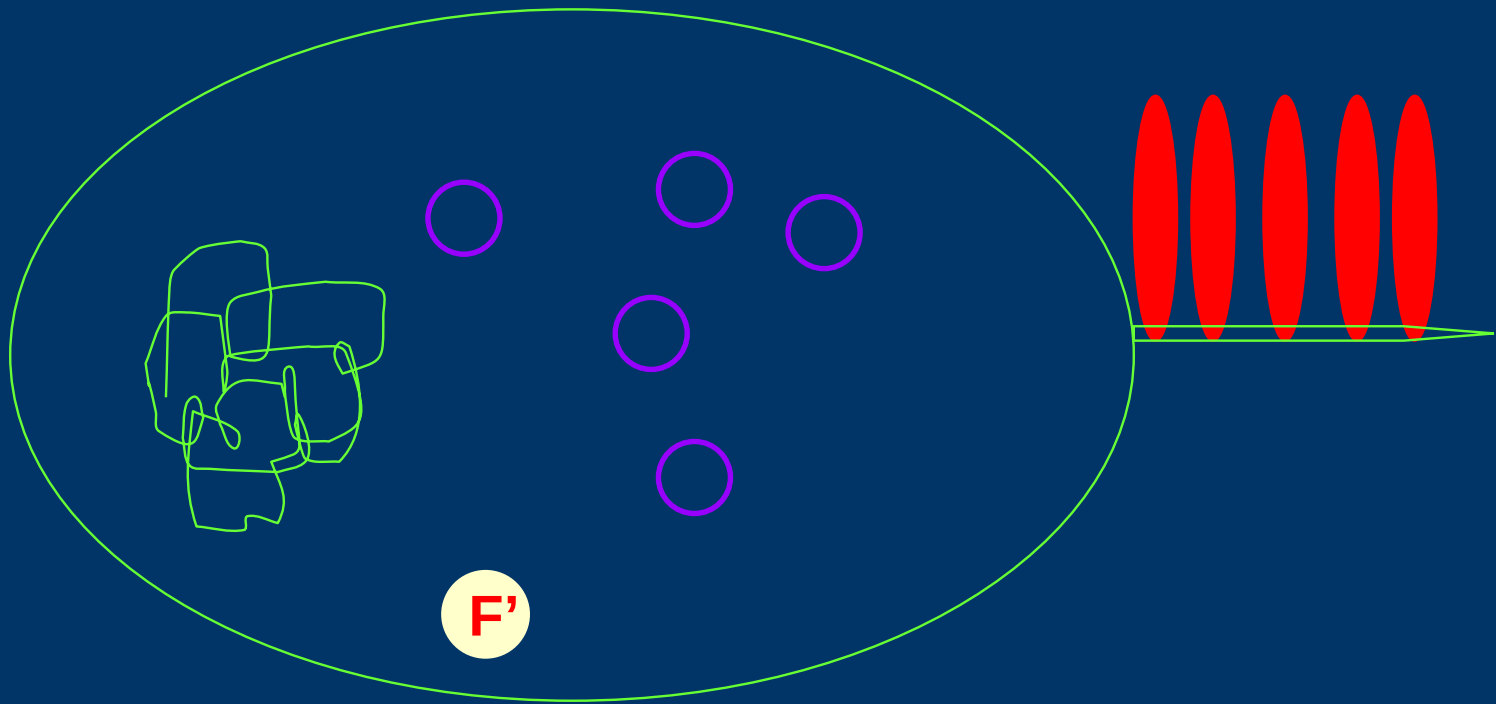
- Head and tail
- Lytic cycle gives clear plaques; host cell will be lysed to release progeny phages; may also enter lysogenic cycle (mixed lysogenic and lytic population give turbid plaques)
- Replicative form is circular and double-stranded
- Packaged DNA are linear and double-stranded

M13, f1

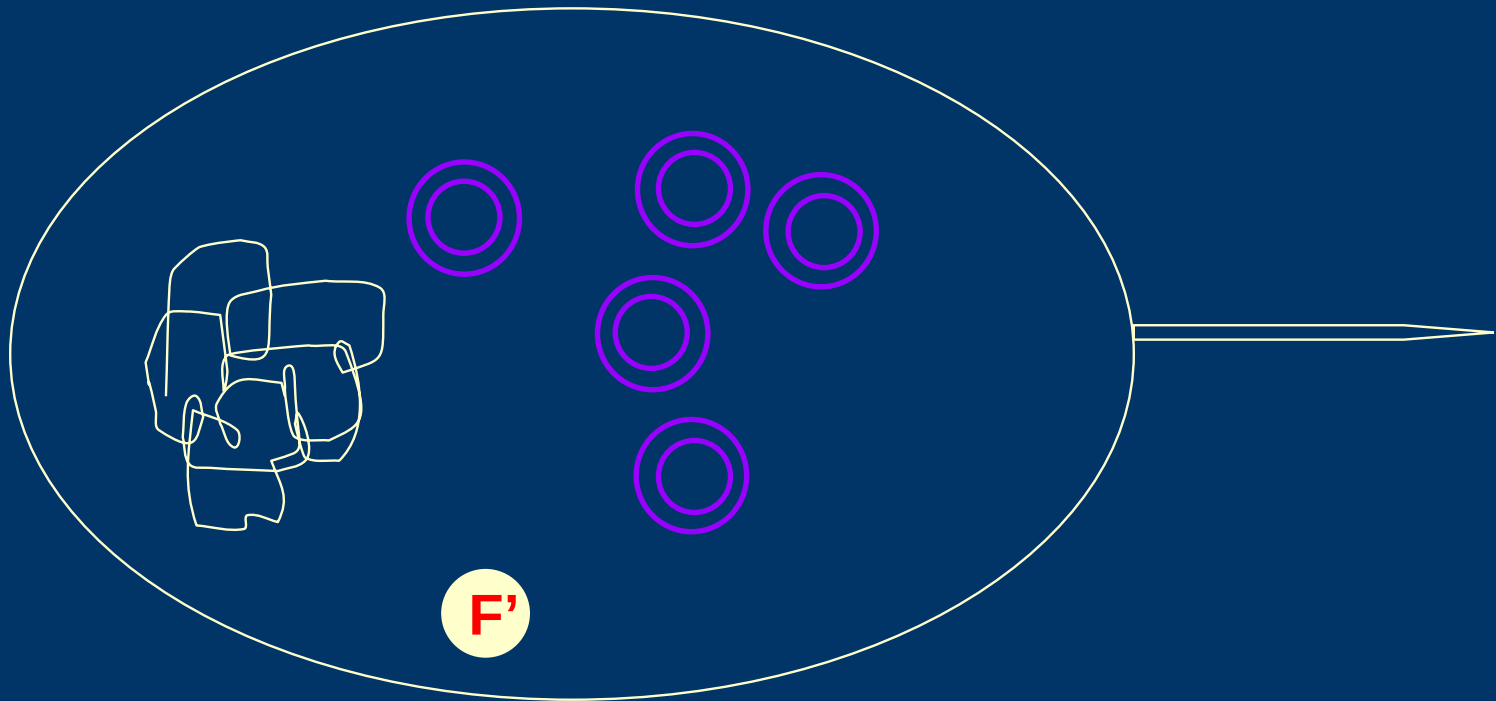
- Filamentous
- No lysis occurs, turgid plaques due to slow growth of infected cells; progeny phages are secreted out of the cells
- Replicative form is circular and double-stranded
- Packaged DNA are circular and single-stranded



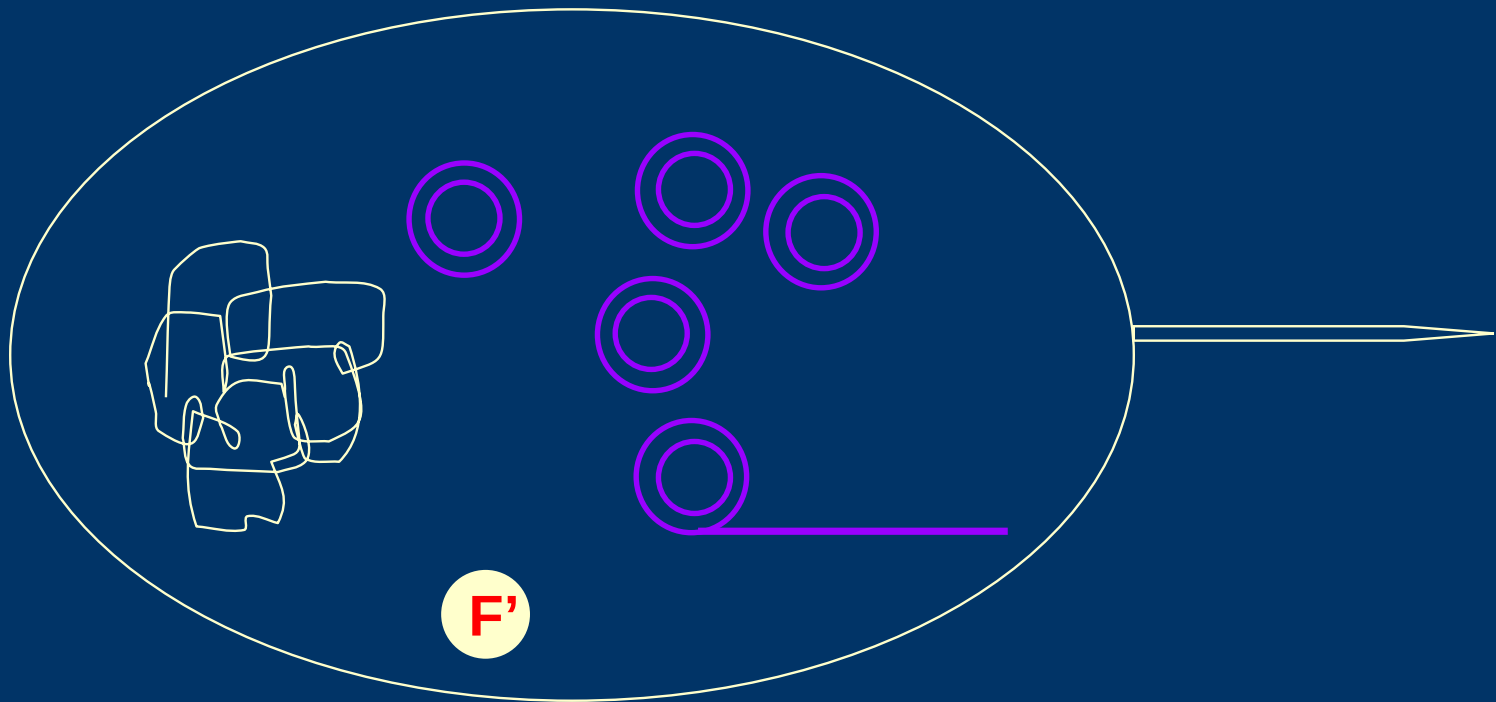
- M13 / f1 phages infect a host cell via F pili



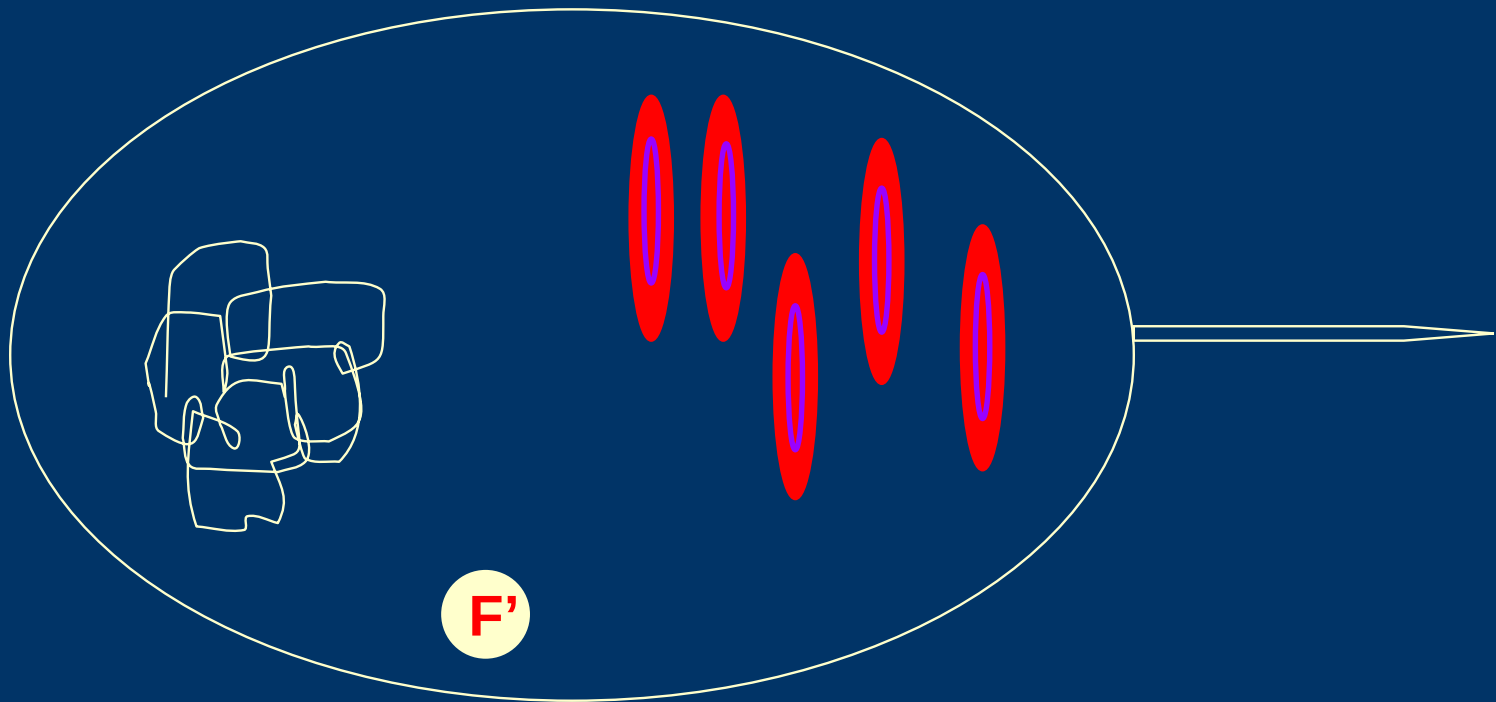
- Phage genome enters the host cell



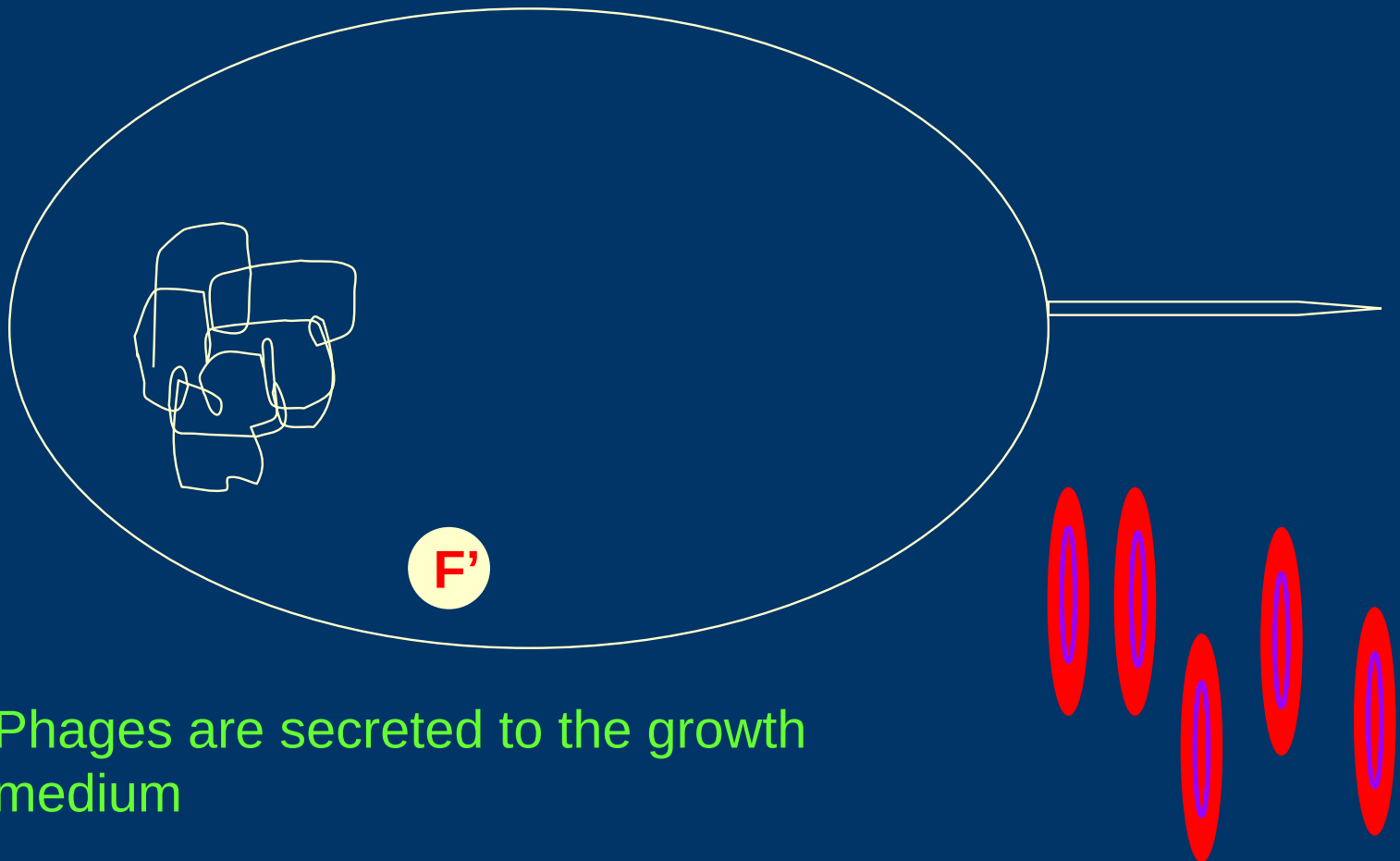
- Formation of double-stranded circular replicative form



- Formation of single-stranded concatemers by rolling circle replication

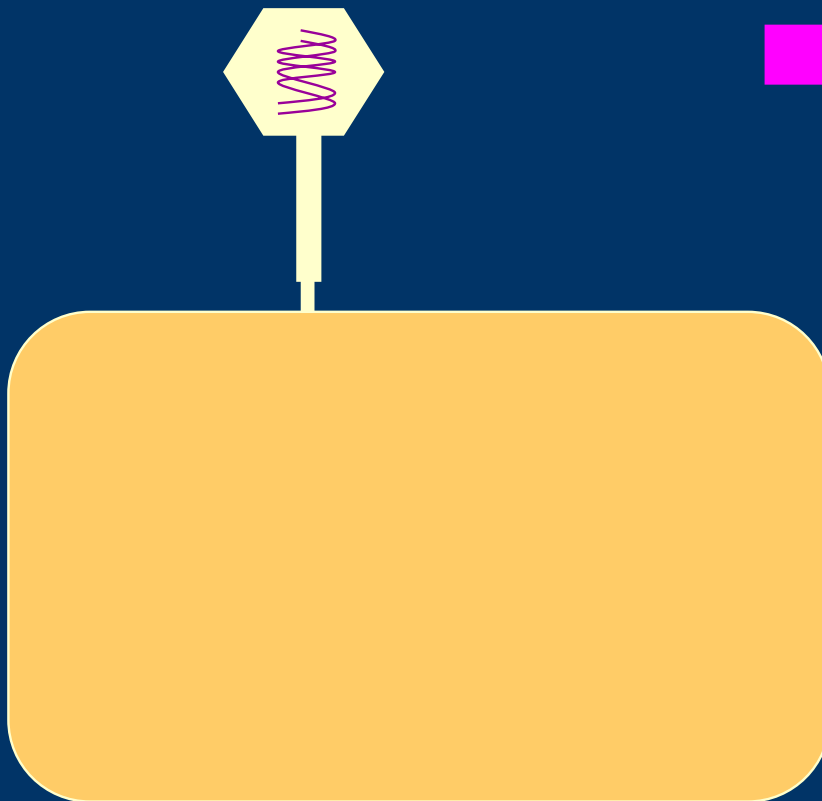


- Single-stranded circularized phage genomes are packaged into phage coats



- Phages are secreted to the growth medium

λ Phage

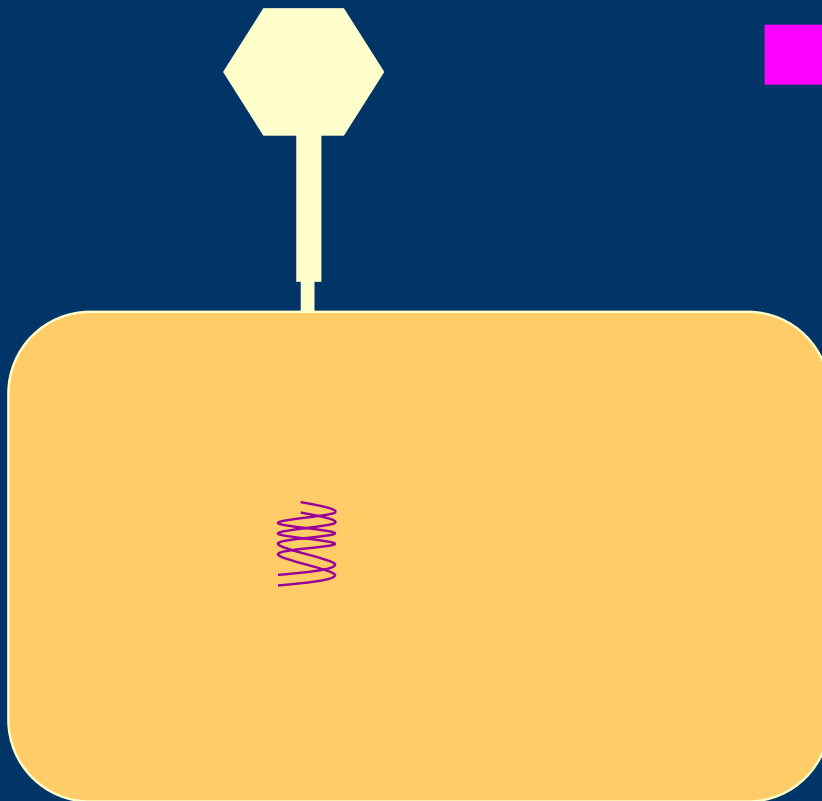


cos



cos

λ Phage



cos



cos

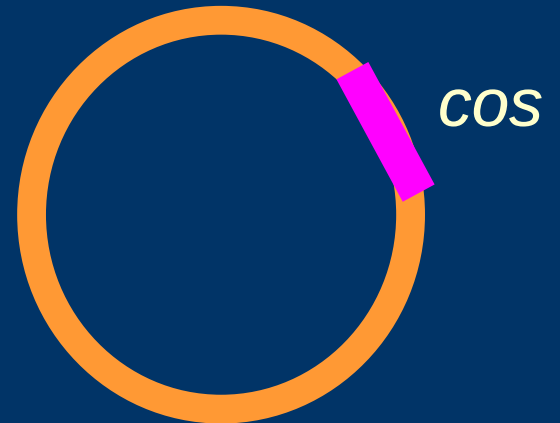
λ Phage

cos

cos



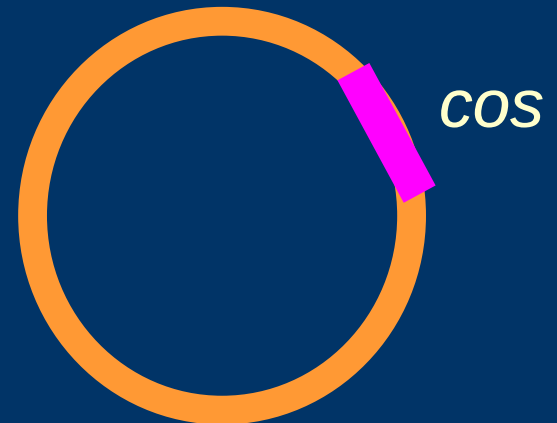
Re-circularized



λ Phage



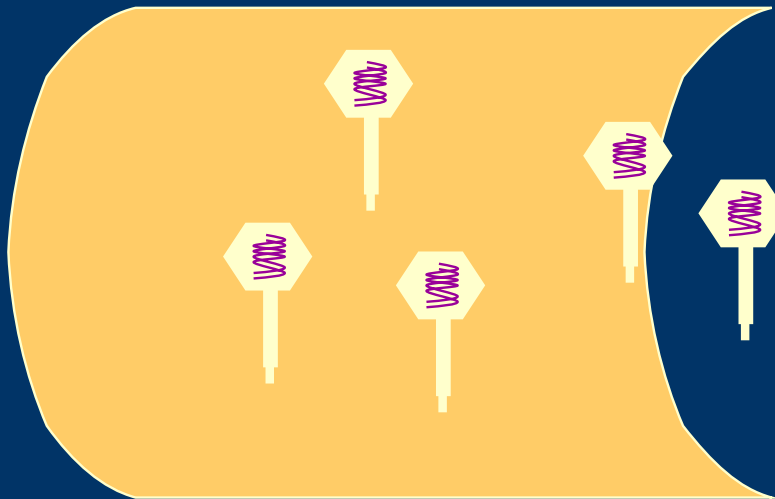
Re-circularized



Formation of concatemers (linear, double-stranded)

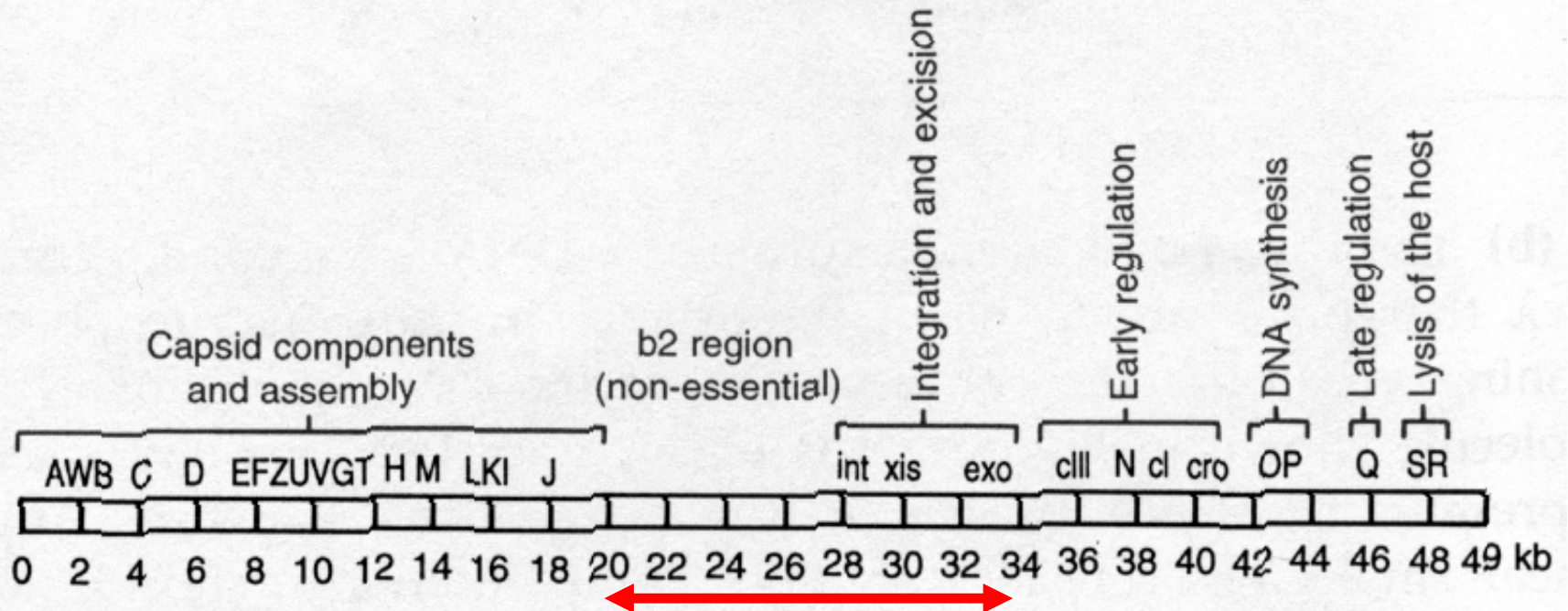


λ Phage



Packaging between two
cos sites; cell lysis

λ Genetic Map



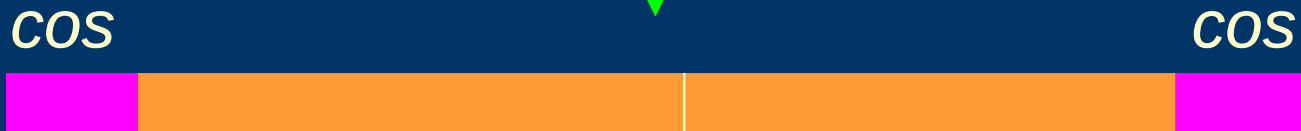
λ Insertion Vectors (e.g. λ gt10, λ ZAP2)



(Total genome size about 47-49 kb)



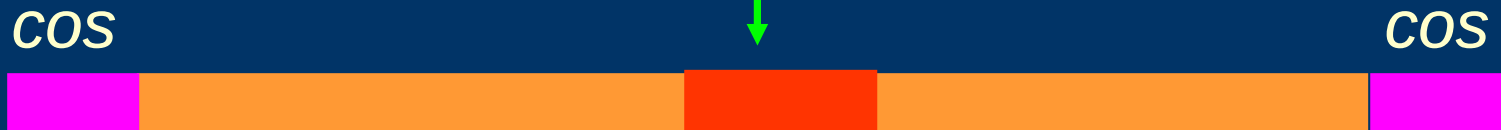
Cleavage and ligation



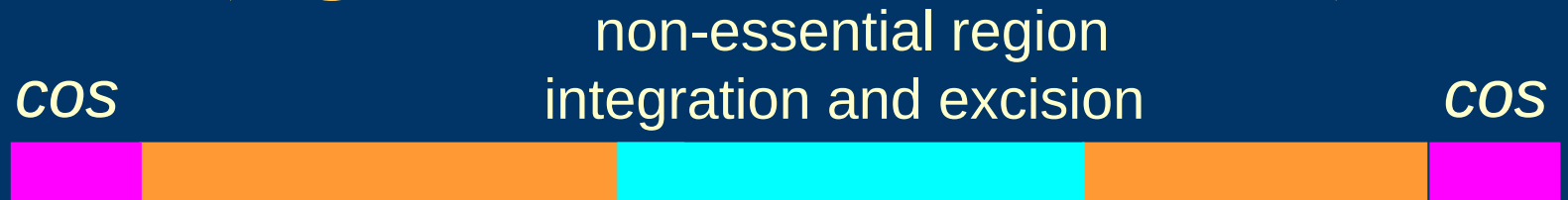
(λ insertion vector size about 38-40 kb)



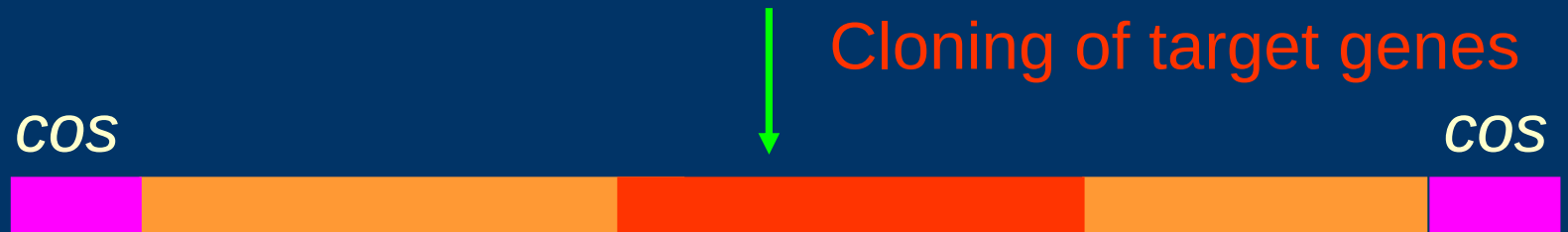
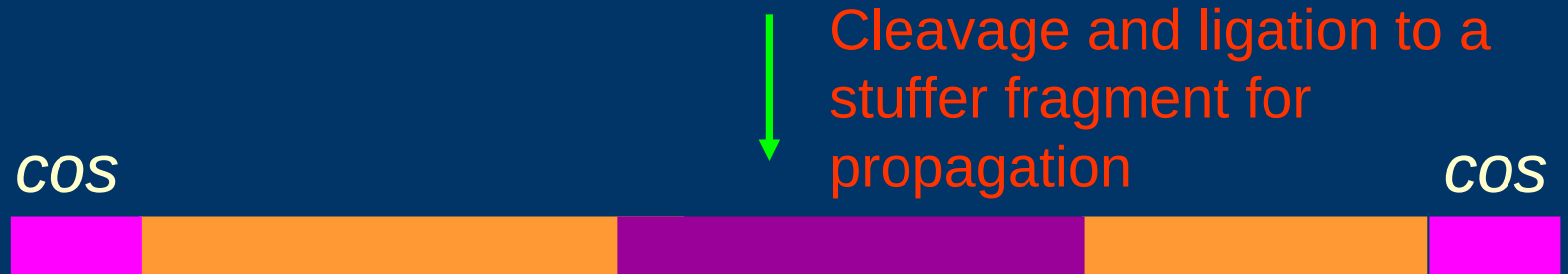
Cloning of target genes



λ Replacement Vectors (e.g. λ WES. λ B', λ EMBL4)



(Total genome size about 47-49 kb)

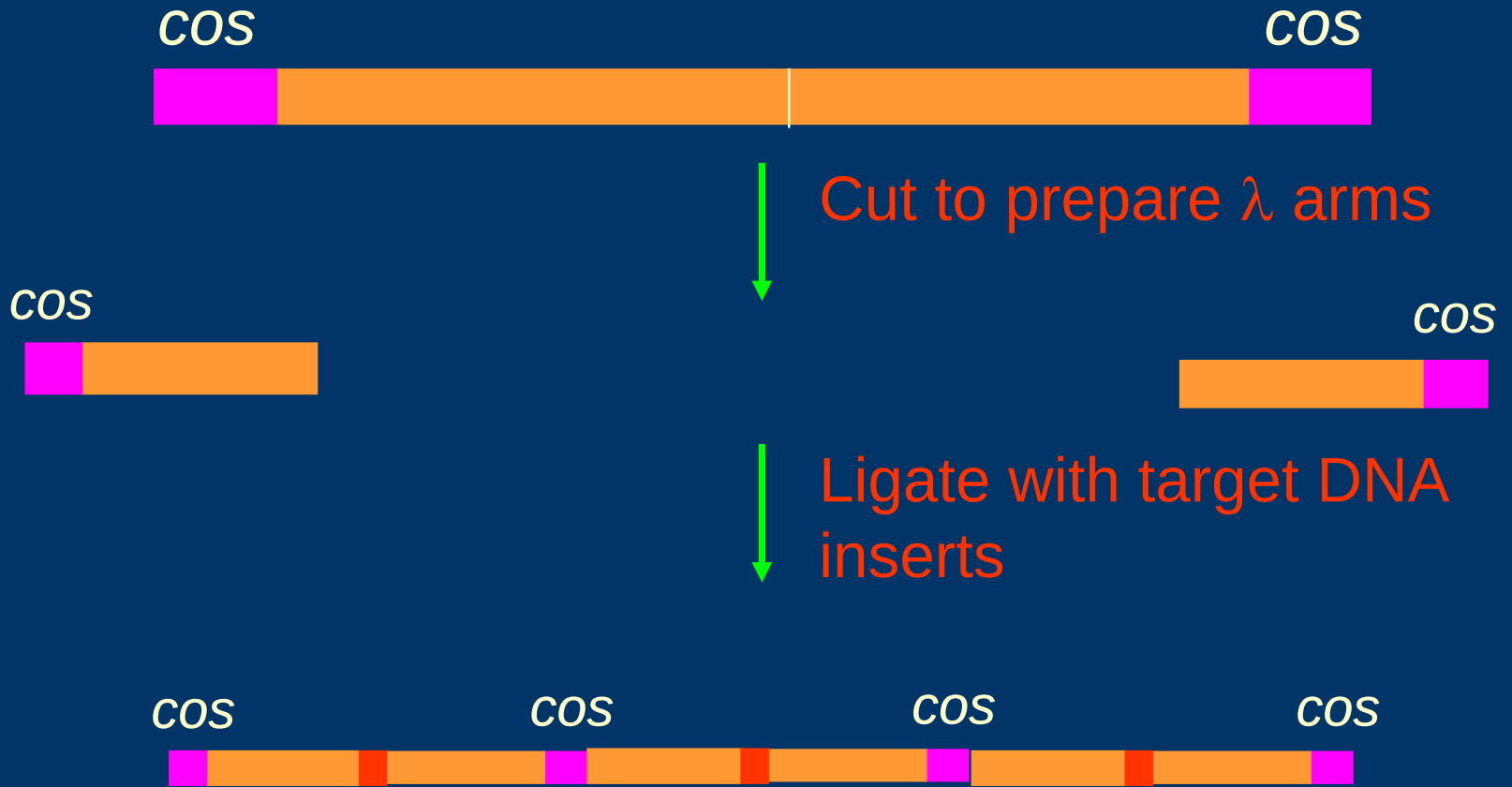


(λ EMBL4 can host DNA inserts with size up to 23 kb)

Cloning with λ DNA

- Use circular form of λ DNA
 - manipulate like a large plasmid
 - transfect into *E. coli*
- Use linear form of λ DNA
 - in form of concatemers

Formation of Concatemers



In Vitro Packaging (Single-Strain System)

- Infect *E. coli* host cells with λ phages defective in the *cos* site
- Proteins for λ phage packaging will be formed but no actual packaging will take place
- Prepare protein extracts from the host cell
- Mix protein extracts with target λ concatemers
- Infect *E. coli* host cells

In Vitro Packaging (Two-Strain System)

- Infect *E. coli* host cells with λ phages defective in making the capsid protein D; no packaging occurs
- Infect *E. coli* host cells with λ phages defective in making the capsid protein E; no packaging occurs
- Prepare protein extracts from the host cells
- Mix these two protein extracts with target λ concatemers
- Infect *E. coli* host cells

Selection of Recombinant Phages

- Cloning site in the *lacZ'* gene
 - recombinant plaque clear, non-recombinant plaques blue in the presence of IPTG and X-gal
 - e.g. λ ZAPII
- Cloning site in the $\lambda cl'$ gene
 - recombinant plaque clear, non-recombinant plaques turbid
 - e.g. λ gt10

Selection of Recombinant Phages

- Cloning site in the *spi* gene
 - recombinant plaque can infect host cells with a P2 prophage, non-recombinant cannot infect;
 - P2 prophage confers immunity to Spi⁺ λ phage
- Selection by size
 - 37-52 kb